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Positive Joint Laplace Inversion: Spectral Support Selection and Identifiability in Diffusion-Ordered Spectroscopy

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Abstract

A small DOSY fitting error does not establish which components are present. We formulate joint positive Laplace inversion for shared diffusion distributions in TRAIn-MF and shared atomic rates in resolution-adaptive inversion (RAI) and diffusion orthogonal moment exchange (DOME). The analysis gives discretization bounds, conditional uniqueness and fixed-rate stability estimates. A common component selector combines residualized kernels, nonnegative variable projection and prediction on reserved acquisitions. Across 36 primary atomic simulations, mean spectral leakage decreases from 5.166% to 1.728% for RAI and from 3.857% to 1.728% for DOME. A factorial experiment supports contributions from both selection stages without establishing an independent advantage of either proposal mechanism. Additional studies include 300 null matrices, 96 weak-component problems, controlled TRAIn-MF convergence tests, five neural training seeds on 120 new problems, classical references and measured DOSY spectra. Weak components are often missed, some TRAIn-MF fits fail the stationarity check, and the neural accuracy gains remain inconclusive. Classical references predict better on the public multicomponent example. These findings separate the demonstrated reduction in spectral leakage from stronger claims about distributional width, chemical identity or confidence after adaptive selection.

Keywords: inverse Laplace transform; positive measures; variable projection; nonnegative least squares; model selection; spectral support; identifiability; DOSY

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1. Introduction

Recovering a positive distribution from a small number of Laplace-transform samples is a prototypical ill-conditioned inverse problem. Diffusion-ordered nuclear magnetic resonance spectroscopy (DOSY) provides a particularly useful setting: the acquisition dimension contains attenuations, while the spectral dimension contains many potentially related signals [1,2]. Different chemical shift positions can carry contributions with the same diffusion coefficient, so independent inversion of every spectral channel discards useful structure. Conversely, imposing excessive sharing can spread a fitted component into spectral regions where it is absent.

Two questions must therefore be separated. The first is how well a positive measure explains the observations. The second is which components, if any, are sufficiently resolved to support an interpretation. A small residual answers neither the second question nor

the question of whether a distribution is atomic or continuous. A sharply drawn DOSY band can be the consequence of the chosen representation, and a smooth band can be the consequence of regularization. Neither appearance alone establishes molecular identity.

Classical approaches address different aspects of Laplace inversion. CONTIN uses constrained regularization [3]; TRAIn uses an iterative trust-region construction [4]; ITAMeD applies iterative thresholding [5]; and PALMA combines entropy and sparsity [6]. Regularized Kaczmarz methods provide an algebraic reconstruction perspective [7]. Joint processing is also established: simultaneous inversion with nonnegative low-rank regularization was investigated by Yuan et al. [8], and high-resolution DOSY reconstruction by Lin et al. [9]. These precedents rule out treating joint inversion, positivity, or a low-rank factorization as novel in themselves.

Applications to protein and polymer molecular-weight estimation motivate reliable diffusion recovery while requiring a separate, condition-dependent calibration [10–12]. Polymerization monitoring, cyclic polymers and chitosan provide related applications [13–16]; they do not validate the present reconstruction benchmark. Algorithmic precedents from this research program include the genetic search in diffGA, algebraic reconstruction in dART, and regularized Kaczmarz inversion [7,17,18]. TRAIn-MF extends the single-channel trust-region method of Xu and Zhang [4] to shared diffusion profiles [19]. Section 4.8 defines the joint objective.

Finite exponential models introduce a second established lineage. Linear amplitudes can be eliminated from nonlinear least squares by variable projection [20]; constrained extensions require attention to active sets and differentiability [21]. Nonnegative least squares (NNLS) supplies the conditional amplitude problem [22]. Continuous sparse-measure methods, including sliding Frank–Wolfe, show that off-grid Laplace support recovery has its own mathematical theory under explicit hypotheses [23]. The method considered here is a finite candidate search with support selection. It is not a new proof of unrestricted super-resolution and does not inherit guarantees from a different estimator.

Our focus is the interface between representation, optimization, and selection. We use a positive measure formulation to compare continuous shared profiles with shared atomic rates. We derive the exact local residual derivative on a stable NNLS active set, explain the information in residualized kernels, and specify a selector that uses spectral coherence without deleting components solely because their heights are small. The selector is applied to two research implementations from the same development program: resolution-adaptive inversion in its component mode (RAI) and diffusion orthogonal moment exchange (DOME). Their different proposal mechanisms feed a common final estimator, denoted RAI-S and DOME-S. The suffix identifies spectral-support selection; it does not designate a new acquisition or a distinct physical model.

The mathematical statements have deliberately limited scope. We prove finite-sampling nonuniqueness for unrestricted positive measures and contrast it with exact uniqueness in a bounded-order noiseless atomic class. We establish a grid-conversion bound, conditional uniqueness of amplitude subproblems, and conditional Gaussian identities motivating selection. These are transparent consequences and adaptations of established principles, rather than claims of priority. The primary comparison shows a reduction in spectral leakage as well as failures in weak and proportional mixtures. For finite-width profiles, TRAIn-MF uses multifrequency matrix factorization. Its factor count need not equal the number of chemical species.

The organization follows this distinction. Section 2 positions the work among related inverse problem methods. Section 3 gives the measure model and mathematical properties. Section 4 formulates each inversion method, including its variables, objective, updates, and selection target. Section 5 derives the common support selector and presents

its structured pseudocode. Appendix D gives additional inversion models. Appendix E establishes a conditional profile uniqueness criterion and covariance-aware identities without claiming their calibration after adaptive selection. Sections 6 and 7 describe the primary benchmark and extended validation. Appendix G reports the new convergence, factorial, weak-signal, multi-seed learning, classical reference and experimental studies. The discussion identifies what remains unresolved, including unknown or correlated noise, structural misspecification and independent chemical validation.

2. Related Inverse Problems and Algorithmic Context

2.1. Representation, Sharing and Regularization

Positive Laplace inversion is a first-kind Fredholm problem in which small singular values limit stable recovery [24,25]. The main methodological distinction is the unknown being estimated. CONTIN and UPEN recover distributions stabilized by smoothing, whereas PALMA combines entropy and sparsity [3,6,27]. Their penalties express different preferences for admissible solutions; none makes finite observations identify an unrestricted positive distribution. The nonnegative smoothing framework of Butler, Reeds and Dawson supplies an earlier reference [26].

SCORE uses shared discrete exponential components, while LOCODOSY fits local spectral windows whose orders can differ [35,37]. These are particularly relevant comparators for the atomic RAI-S and DOME-S models. Global sharing pools evidence across resonances; local fitting limits how much unrelated spectral structure is coupled. Neither choice removes the need to determine which components contribute at each chemical shift position. OUTSCORE provides a further component-resolution approach, and GNAT makes these distinctions explicit in a common processing environment [36,38]. Multivariate curve resolution and nonnegative matrix factorization provide broader precedents for shared-factor models [33,34]. In TRAIn-MF, the attenuation factor is additionally constrained to KS with a known kernel K. Normalizing the profiles removes a scale freedom but not all factorization ambiguity; a profile can remain broad or multimodal.

Table 1 contrasts seven relevant external methods across distributional, component-based and learned representations. It identifies feasible comparisons and their contracts. The frozen primary benchmark is supplemented by the executed SCORE, LOCODOSY and PALMA-objective comparisons in Appendix G.5; the remaining entries are context rather than executed rankings.

Table 1. Properties and requirements for external comparisons. Availability of source code or trained weights is distinct from validation on a common dataset.

Method	Representation and Stabilizing Information	Requirement for a Fair DOSY Comparison
CONTIN [3]	Nonnegative distribution; constrained smoothing	Match kernel, grid convention and regularization selection; compare prediction and distributional error.
PALMA [6]	Positive distribution; entropy and sparsity	Fix the penalty mixture, noise constraint and stopping rule without test-data tuning.

Table 1. *Cont.*

Method	Representation and Stabilizing Information	Requirement for a Fair DOSY Comparison
UPEN [27]	Distribution; locally adapted penalty	Use a verified one-dimensional Laplace implementation, not a second inversion over chemical shift.
SCORE [35]	Shared discrete components and spectral amplitudes	Declare order selection, initialization, amplitude constraints and computation budget.
LOCODOSY [37]	Local multivariate fits and local component orders	Derive windows and orders from training data; distinguish local contributions from global species.
DRECT [41]	Learned Laplace-NMR reconstruction	Match acquisition sampling, units, normalization and training domain; test distribution shift.
DREAM [42]	Learned reconstruction with uncertainty estimates	Match the same input contracts and assess uncertainty calibration on independent data.

The code is available for these methods or related implementations, and pretrained DOSY checkpoints are available for DRECT and DREAM. This makes direct comparisons feasible but does not make their interfaces interchangeable. Distributional methods do not natively estimate an atomic order; LOCODOSY's local component count is not a global species count; and a learned model may require sampling or normalization different from the present protocol. Adapters must preserve physical units and the observations used for evaluation. In particular, two-dimensional relaxation tools [30–32] cannot treat the chemical shift axis as a second Laplace dimension. The primary frozen benchmark does not establish a ranking against these seven external methods. The additional executed comparisons concern the explicitly attributed implementations, adapters, parameter rules and endpoints in Appendix G.5; CONTIN, UPEN, DRECT and DREAM remain unexecuted in that common protocol.

2.2. Optimization, Identification and Learning

Efficient optimization does not overcome a lack of information. FLINT addresses the computational cost of distribution estimation, and FISTA supplies convergence results for specified convex objectives [28,29]; those results do not automatically cover a bilinear factorization or adaptive support and order changes. ESPIRA estimates exponential parameters through rational approximation and structured matrices [39]. Its sampling assumptions, like the assumptions of any identification theorem, must be checked for the model under consideration [40]. Noiseless uniqueness in a bounded-order atomic class and stable recovery of nearby rates in noise are different claims.

DRECT and DREAM also obtain information from their training distributions [41,42]. Their evaluation therefore requires independent test cases and checks under distribution shift. RAI-Net has a different role: it changes the order of refinement proposals in an adaptive solver. Neither the term "learning" nor use of a physical kernel makes it equivalent to a physics-informed neural network, and the present selector's conditional Gaussian calculations do not provide DREAM's learned uncertainty model or a calibrated chemical confidence interval.

The distinction between representation, optimization and evidence extends beyond Laplace NMR. Fractional-diffusion studies compare optimizers for recovering derivative orders and boundary conditions [43]; hybrid searches have been investigated for reconstruction from incomplete tomographic measurements [44]. Studies of physics-based neural networks further motivate sensitivity analyses across domain geometry and training hy-

perparameters [45]. These examples involve different forward operators and unknowns: a fractional partial differential equation, tomographic projection data, and differential-equation solutions, respectively. They provide methodological context, not direct DOSY performance comparators or validation of RAI-Net.

3. Positive Measures, Discretization, and Identifiability

3.1. Observation Model and the Meaning of a Diffusion Distribution

Let $I_D = [D_{\min}, D_{\max}] \subset (0, \infty)$ and let μ_ℓ be a finite nonnegative Borel measure for spectral channel $\ell = 1, \ldots, p$. With acquisition coordinates $b_i \geq 0$, $i = 1, \ldots, n$, write

$$Y_{i\ell} = (\mathcal{L}\mu_\ell)_i + \varepsilon_{i\ell}, \qquad (\mathcal{L}\mu)_i = \int_{I_D} e^{-b_i D}\, d\mu(D). \tag{1}$$

Here, b_i has units $\mathrm{s\,m^{-2}}$ and D has units $\mathrm{m^2\,s^{-1}}$. For ideal rectangular gradient pulses, $b = \gamma^2 g^2 \delta^2 (\Delta - \delta/3)$ [1]; other waveforms require their own calibrated encoding factor. Merely writing $b = uv$ does not add independent information when the kernel remains e^{-Duv}. We consider pure diffusion, without exchange, convection, relaxation contrast, or an estimated baseline. Those effects require a different observation model.

The benchmark assumes independent $N(0, \sigma^2)$ errors with known σ. Real signed absorption data are permitted: positivity constrains the reconstructed measure, not every noisy datum. Taking absolute values or clipping negative observations would change this noise model. Spectral channels outside a declared signal mask are unestimated, rather than inferred to contain zero chemical signal.

Let $D_0 > 0$ be a reference diffusion coefficient and $x = \log(D/D_0)$, with $I_x = [a, c]$. The push-forward measure ν obeys

$$(\mathcal{L}\mu)_i = \int_{I_x} k_i(x)\, d\nu(x), \qquad k_i(x) = e^{-b_i D_0 e^x}. \tag{2}$$

If $d\mu = f(D)dD = \rho(x)dx$, then $\rho(x) = D_0 e^x f(D_0 e^x)$. A density per unit D, a density per unit $\log D$, and a bin mass are different quantities. In particular, the height of a histogram representing an atom diverges as the bin width tends to zero, while its mass remains fixed.

3.2. Mass-Preserving Discretization and a Display Error Bound

Let I_x be partitioned into cells C_h of width at most δ. Bin masses are $m_h = \nu(C_h)$; plotting cell densities means displaying $m_h/|C_h|$. If $Q\nu = \sum_h m_h \delta_{x_h}$, with x_h the midpoint of C_h, then total mass is preserved exactly. This operation does not transform an atom into a physically broadened component.

Proposition 1 (Quantization and forward error)**.** *For a positive measure ν of mass $M > 0$, midpoint quantization satisfies*

$$W_1(\nu/M, Q\nu/M) \leq \delta/2, \qquad |\mathcal{L}_i\nu - \mathcal{L}_i Q\nu| \leq M\delta/(2e). \tag{3}$$

For two normalized measures ν, η, both quantized on such a grid, $|W_1(\nu, \eta) - W_1(Q\nu, Q\eta)| \leq \delta$.

Proof. Transport each point to its cell midpoint. Every displacement is at most $\delta/2$, giving the first bound. Since $|k_i'(x)| = te^{-t} \leq e^{-1}$ for $t = b_i D_0 e^x \geq 0$, integration of the Lipschitz bound gives the second. The final statement follows twice from the triangle inequality for W_1. □

This is an approximation bound, not a resolution theorem. The atomic solver below evaluates its exponentials at the fitted continuous rates, not at histogram midpoints. Its 256-bin export is used for compatible visualization and a declared binned evaluation. On $D/D_0 \in [0.1, 15]$, $\delta = \log(150)/256 \simeq 0.019573$; native and binned transport comparisons need not agree below this scale. The one-dimensional formula $W_1(\nu, \eta) = \int |F_\nu - F_\eta| dx$ is classical [46]. Normalized shape is undefined for a zero-mass measure and is treated separately below.

3.3. Finite Observations and Restricted Atomic Uniqueness

Proposition 2 (Ambiguity of unrestricted positive measures)**.** *For any n finite Laplace sampling coordinates and any interval with nonempty interior, there exist two distinct positive measures with the same total mass and identical sampled transforms.*

Proof. Choose $n+2$ distinct points D_j in the interval and form the $(n+1) \times (n+2)$ matrix whose first row is one and whose other rows are $e^{-b_i D_j}$. A nonzero null vector v exists. Since $\sum_j v_j = 0$, it has both positive and negative entries. The measures $\mu_+ = \sum_{v_j > 0} v_j \delta_{D_j}$ and $\mu_- = \sum_{v_j < 0} (-v_j) \delta_{D_j}$ are positive, nonzero, distinct, and have equal masses and transforms by construction. □

The proposition is an existence statement about the unrestricted class; it does not say that every positive measure is ambiguous under every finite-dimensional restriction. In particular, the following elementary atomic result prevents an overstatement of impossibility.

Proposition 3 (Noiseless uniqueness with a prescribed order bound)**.** *On a compact positive interval, samples at $b_i = b_0 + ih$, $i = 0, \ldots, 2r-1$, with $h > 0$, distinguish any two measures having at most r atoms.*

Proof. The difference of two such measures has at most $s \leq 2r$ distinct support points. Absorb $e^{-b_0 D_j}$ into their signed weights and put $z_j = e^{-hD_j}$. Equality of the first s samples gives $\sum_{j=1}^{s} c_j z_j^i = 0$, $i = 0, \ldots, s-1$. The resulting square Vandermonde matrix is nonsingular because the z_j are distinct, so every weight vanishes. □

No stability constant uniform over colliding rates follows from this proof. At finite noise, nearby exponentials can be almost indistinguishable. If two components have exactly the same D, only their summed spectral amplitudes are observed in Equation (1); diffusion alone cannot separate them into distinct species.

For a quantitative statement with *fixed correct rates*, normalize the kernel columns: $B = K \operatorname{diag}(\|k_j\|_2)^{-1}$ and $\beta = \operatorname{diag}(\|k_j\|_2) A$. Let $\mu = \max_{j \neq k} |B_j^\top B_k|$ for $r \geq 2$, and set $\mu = 0$ for $r = 1$. If $\gamma = 1 - (r-1)\mu > 0$, Gershgorin's bound gives $\sigma_{\min}(B) \geq \sqrt{\gamma}$. For $Y = B\beta + E$, $\beta \geq 0$, and its NNLS estimate $\widehat{\beta}$,

$$\|\widehat{\beta} - \beta\|_F \leq \frac{\|E\|_F}{\sigma_{\min}(B)} \leq \frac{\|E\|_F}{\sqrt{1-(r-1)\mu}}.$$

Indeed, the fitted matrix is the Euclidean projection onto the closed cone generated by B; nonexpansiveness gives $\|B(\widehat{\beta} - \beta)\|_F \leq \|E\|_F$. The smallest singular value then gives the first inequality. This is an amplitude stability bound for fixed rates, not a rate estimation bound. The coherence criterion is sufficient and conservative: when $\gamma \leq 0$, the actual $\sigma_{\min}(B)$ may still be positive.

For two normalized columns, $\sigma_{\min}(B)^2 = 1 - \rho$, where ρ is their correlation. Put $k_i(D) = e^{-b_i D}$, $w_i(D) = k_i(D)^2/\|k(D)\|_2^2$, $\bar{b}_w = \sum_i w_i b_i$, and $P_k = kk^\top/\|k\|_2^2$. For a fixed acquisition design and small physical rate difference δ,

$$1 - \rho(D, D+\delta) = \tfrac{1}{2}\delta^2 \operatorname{Var}_{w(D)}(b) + O(\delta^3),$$
$$\|(I - P_{k(D)})k(D+\delta)\|_2 = \|k(D)\|_2 |\delta| \sqrt{\operatorname{Var}_{w(D)}(b)} + O(\delta^2).$$

These expansions follow by differentiating $k/\|k\|_2$: its ith derivative is $(\bar{b}_w - b_i)k_i/\|k\|_2$. Thus, conditioning depends on the sampled b range as well as rate separation. Under the fixed-rate, fixed-group assumptions of Proposition 7, equal amplitude a over m independent channels gives detection power

$$\Pr(Z_{jG} > z_\alpha) = \Phi\left(\frac{a\|v_j\|_2\sqrt{m}}{\sigma} - z_\alpha\right), \qquad z_\alpha = \Phi^{-1}(1 - \alpha/(rg)).$$

Consequently, power at least $1 - \beta_{\text{miss}}$ requires $a\|v_j\|_2\sqrt{m}/\sigma \geq z_\alpha + \Phi^{-1}(1 - \beta_{\text{miss}})$. The two-rate expansion gives a local separation scale proportional to

$$\frac{\sigma}{a\|k(D)\|_2\sqrt{m \operatorname{Var}_{w(D)}(b)}}.$$

This is a design- and SNR-dependent calculation, not a universal minimum gap or a calibrated resolution guarantee for adaptively estimated rates and groups.

For an atomic model, $Y^0 = K(D)A$, where $K_{ij} = e^{-b_i D_j}$ and $A \geq 0$. If K has a full column rank, $\operatorname{rank}(Y^0) = \operatorname{rank}(A)$. When $A = wc^\top$ is proportional across spectral channels, the clean matrix has rank one even if w has several nonzero entries. The common decay can still be multiexponential and, under Proposition 3's assumptions, identifiable from noiseless samples. Proportionality removes spectral diversity; it is not by itself a proof of scalar exponential nonidentifiability. Repeated noisy versions can improve precision without increasing matrix rank.

Proposition 4 (What a noise threshold says about rank)**.** *If $Y = Y^0 + E$ and $\|E\|_2 \leq \tau$, the number of singular values of Y exceeding τ is no greater than* $\operatorname{rank}(Y^0)$.

Proof. For $k > \operatorname{rank}(Y^0)$, the singular-value perturbation inequality gives $s_k(Y) \leq s_k(Y^0) + \|E\|_2 \leq \tau$. □

Thus, resolved matrix directions provide a lower count, not an upper bound on hidden rates or chemical species. A search capped at this count makes an additional predictive modeling choice. This matters for the archived automatic TRAIn-MF procedure, whose cap targets resolved factors. It must not be interpreted as a consistent chemical order estimator.

3.4. Distributional Sharing, Normalization, and Partial Sharing

On a mass grid, a shared distributional model is $X = SA$, with

$$S \in \mathbb{R}_+^{q\times r}, \quad A \in \mathbb{R}_+^{r\times p}, \quad \mathbf{1}^\top S = \mathbf{1}^\top. \tag{4}$$

The simplex constraint fixes the scaling ambiguity $S_{\cdot j} \mapsto t_j S_{\cdot j}$, $A_{j\cdot} \mapsto A_{j\cdot}/t_j$. It does not fix permutation ambiguity, coincident profiles, or nonunique factorizations. Each column of S is a distribution and may have several modes; its index is not necessarily a species label.

A partially shared model is $C = SA + V$, $V \geq 0$, on nonnegative unit-integral basis functions ϕ_h. Then $\rho_\ell(x) = \sum_h C_{h\ell}\phi_h(x)$ and its total mass is $\sum_j A_{j\ell} + \mathbf{1}^\top V_{\cdot\ell}$. Because $A = 0, V = C$ is feasible for every $C \geq 0$, the private term preserves the full positive basis cone. The shared rank changes the representation cost rather than the set of attainable total densities. The decomposition itself is not generally identifiable: for $c_\ell = m_\ell s$, every $A_{1\ell} = tm_\ell, V_{\cdot\ell} = (1-t)m_\ell s, 0 \leq t \leq 1$, produces the same density.

One may regularize this model with

$$J = \frac{1}{2p}\Big[\|\sigma^{-1}(KC - Y)\|_F^2 + \lambda_s \operatorname{tr}(C^\top QC) + \lambda_p \operatorname{tr}(V^\top GV)\Big], \tag{5}$$

where $K_{ih} = \int k_i\phi_h$, $G_{hk} = \int \phi_h\phi_k$, and $Q_{hk} = \int \phi_h'\phi_k'$. The penalties act on the total-density derivative and the private density, respectively. They are not interchangeable.

Proposition 5 (Conditional uniqueness for one shared profile)**.** *For a fixed unit-mass profile s, let $L^\top L = Q$, $F^\top F = G$, and $T = [K/\sigma; \sqrt{\lambda_s}L]$. If G is positive definite, $\lambda_p > 0$, and $Ks \neq 0$, the conditional NNLS problem for (a_ℓ, v_ℓ) has a unique solution.*

Proof. Its design matrix is

$$B(s) = \begin{bmatrix} Ts & T \\ 0 & \sqrt{\lambda_p}F \end{bmatrix}, \qquad d_\ell = \begin{bmatrix} y_\ell/\sigma \\ 0 \\ 0 \end{bmatrix}. \tag{6}$$

If $B(s)(\alpha, z)^\top = 0$, the lower block gives $Fz = 0$, hence $z = 0$. Then, $Ts\alpha = 0$ implies $\alpha = 0$. Full column rank makes the quadratic objective strictly convex, giving a unique constrained minimizer. □

The uniqueness result applies to the conditional subproblem. It does not imply a unique outer profile or molecular decomposition. The partial-sharing model is not included in the primary atomic comparison.

4. Method-Specific Inversion Problems

4.1. Notation, Representation, and Comparison Scope

Let $Y \in \mathbb{R}^{n\times p}$ contain the retained spectral channels, with acquisition coordinates b_i, and let $\Sigma = (\sigma_{i\ell})$ contain their noise standard deviations. Division by Σ below is entrywise; the benchmark uses the common value σ. We distinguish an atomic kernel $K(d)_{ij} = \exp(-b_i d_j)$ from an integrated basis operator

$$B_{ih} = \int_{x_{\min}}^{x_{\max}} e^{-b_i D_0 e^x}\varphi_h(x)\,dx, \qquad \varphi_h \geq 0, \quad \int \varphi_h(x)\,dx = 1. \tag{7}$$

For the latter, $C_{h\ell} \geq 0$ is mass, not density height, and BC predicts the observations. Integration and discretization are part of the forward model, not a cosmetic change to the DOSY display. In contrast, exporting continuous atomic rates to 256 display cells does not turn the atomic fit into a 256-parameter inversion.

Table 2 distinguishes the unknowns and selection targets of the methods. In the primary comparison, RAI denotes its finite-component mode. Density RAI, adaptive Haar RAI and RAI-Net solve different estimation problems. Appendix D describes RAI-FLEX, partial-C, CIRCE and support averaging. Appendix C and Supplementary File S1 give the computational details and study protocols.

Table 2. Mathematical contracts. A selected representation size is not, in general, a chemical species count.

Method	Unknowns and Prediction	Coupling or Penalty	Selection Target
TRAIn	$h_\ell \geq 0$, Kh_ℓ	Independent columns; iterative stopping	Stopping iterate
RAI, atomic	$d, A \geq 0$, $K(d)A$	Shared rates	Native BIC-type order
RAI, density	$C \geq 0$, BC	Roughness, ridge, spectral graph	Grid and validation error
Adaptive RAI / RAI-Net	$U \geq 0$, $B_{\mathcal{T}}U$	Shared Haar details	Tree and penalty; learned action order for Net
DOME	$d, A \geq 0$, $K(d)A$	Shared rates, split/exchange proposals	Native corrected score
RAI-S / DOME-S	d, A, H, $K(d)A$	Allowed spectral support H	Predictive order and support
TRAIn-MF	$S, A \geq 0$, KSA	Simplex profiles, curvature and ridge	Resolved predictive factors

4.2. TRAIn: Squared Positivity and a Trust-Region Iteration

The supplied single-channel implementation uses a TRAIn-derived trust-region Gauss–Newton iteration [4]. Its numerical core minimizes the unpenalized residual through the squared parametrization

$$h = \eta \odot \eta, \qquad f(\eta) = \|Kh - y\|_2^2, \quad g = 4\,\mathrm{diag}(\eta)K^\top(Kh - y). \tag{8}$$

The Gauss–Newton matrix and trust-region subproblem are

$$B_{\mathrm{GN}} = 8\,\mathrm{diag}(\eta)K^\top K\,\mathrm{diag}(\eta), \qquad \min_{\|s\|_2 \leq \Delta} g^\top s + \tfrac{1}{2}s^\top B_{\mathrm{GN}} s. \tag{9}$$

The exact Hessian additionally contains $4\,\mathrm{diag}(K^\top(Kh - y))$; the code deliberately uses the Gauss–Newton approximation. Truncated conjugate gradients approximate Equation (9), stopping at the radius boundary or on nonpositive curvature, with at most 100 inner iterations. The trial point is $\eta^+ = |\eta + s|$, which leaves its squared physical prediction unchanged. With a positive predicted reduction, the acceptance ratio is

$$\rho = \frac{f(\eta) - f(\eta^+)}{-g^\top s - s^\top B_{\mathrm{GN}} s/2}. \tag{10}$$

The supplied implementation accepts when $\rho > 0.01$, halves the radius when $\rho \leq 0.25$, and doubles it when $\rho \geq 0.75$, subject to its initial radius cap. These are the settings of the inspected file, not a claim that every published TRAIn implementation has identical defaults.

First, it computes $h_{\mathrm{NNLS}} = \arg\min_{h \geq 0} \|Kh - y\|_2$ and $\nu = \|Kh_{\mathrm{NNLS}} - y\|_2$. Iteration stops when $\sqrt{f} \leq \texttt{termFac}\,\nu$ or the iteration budget is exhausted. Thus, ν is a residual floor from the same data, not an independently measured noise level. Despite a Tikhonov description in the interface, this core contains no explicit term $\lambda\|Lh\|^2$: its regularizing effects come from the parametrization, initialization, and stopping. Its final threshold $h_j < 10^{-6}\max_k h_k$ is a further output modification.

Squared positivity also changes stationarity: $\eta = 0$ gives $g = 0$ even when $K^\top(Kh - y)$ has negative entries and the NNLS KKT conditions fail. Indeed, the supplied initialization

is proportional to $\min_i |y_i|$ and becomes zero if that minimum is zero. Consequently, small g is not by itself a certificate of NNLS optimality. Independent calls at each spectral channel minimize a separable sum of these residuals; they impose no shared components across spectral channels.

4.3. RAI in Finite-Component Mode

For each candidate order r, RAI profiles out the amplitudes:

$$\psi_r(z) = \min_{A \geq 0} \frac{1}{2} \|(K(D_0 e^z)A - Y)/\Sigma\|_F^2, \quad z \in [\log(D_{\min}/D_0), \log(D_{\max}/D_0)]^r. \tag{11}$$

Here, $D_0 = \sqrt{D_{\min} D_{\max}}$ in the component implementation. At each evaluation, SciPy NNLS solves the weighted amplitude problem separately for each spectral channel. Dividing a data column and its noise by the same positive scale, and rescaling its amplitude variable accordingly, leaves Equation (11) unchanged. The native code uses $s_\ell = \max(\max_i |Y_{i\ell}|, \text{median}_i\, \sigma_{i\ell})$ for this conditioning step.

On a stable NNLS active set, the envelope gradient is

$$\frac{\partial \psi_r}{\partial z_j} = \sum_\ell A_{j\ell} (\partial_{z_j} k_j)^\top \left((KA - Y)_\ell / \Sigma_\ell^2\right), \qquad (\partial_{z_j} k_j)_i = -b_i d_j e^{-b_i d_j}. \tag{12}$$

Bounded L-BFGS-B updates the log rates. This native RAI step is not the squared-amplitude TRAIn trust-region iteration. Ten starts combine an interior equispaced grid, up to three births chosen by residual improvement on a 49-point log-rate grid extending the preceding order and seeded random starts. The lowest attained objective determines the candidate, with a nearly tied converged start preferred within the implementation's objective tolerance. This is a finite multistart search, not a global minimization theorem.

The native selector includes $r = 0$ and minimizes

$$\text{BIC}_{\text{RAI}}(r) = 2\psi_r + r(p+1)\log(np). \tag{13}$$

It counts every amplitude, including zeros. The primary comparison caps r at four. This score is a heuristic adaptation of BIC [47]: inactive amplitudes and coincident rates make the mixture singular, invalidating a direct appeal to regular-model dimension counting [48]. Native optimization statuses are saved; the native selection itself does not enforce the stronger eligibility checks subsequently imposed by RAI-S. There is no spectral graph or density smoothing penalty in this atomic RAI objective.

4.4. RAI with a Positive Density and a Spectral Graph

Let log bins have widths w_h and centers c_h, and use the unit-integral indicator bases in Equation (7). The operator is evaluated by 12-point Gauss–Legendre quadrature per cell. Let $H_w = \text{diag}(1/w_h)$ and define the density-difference operator by

$$(RC)_{h\ell} = \frac{C_{h+1,\ell}/w_{h+1} - C_{h\ell}/w_h}{\sqrt{c_{h+1} - c_h}}. \tag{14}$$

In the solver's column-normalized units, the fixed-grid, fixed-graph problem is

$$\min_{C \geq 0} \frac{\|(BC - Y)/\Sigma\|_F^2}{2n} + \frac{\lambda_s}{2}\|RC\|_F^2 + \frac{\eta}{2}\,\text{tr}(C^\top H_w C) + \frac{\gamma}{2}\,\text{tr}(H_w C L_g C^\top). \tag{15}$$

Here $L_g = \text{diag}(W\mathbf{1}) - W$ is a symmetric positive semidefinite graph Laplacian on spectral channels. The identity $\text{tr}(H_w C L_g C^\top) = \sum_{\ell<m} W_{\ell m} \|H_w^{1/2}(C_\ell - C_m)\|_2^2$ shows that this penalty rewards similar density shapes at connected spectral channels. It is not

a chemical assignment constraint. Unlike the unregularized atomic objective, column normalization here also defines the relative scale of the density prior; physical masses are restored after fitting.

The graph is constructed from normalized training decays, retaining sufficiently informative signals, mutual nearest neighbors, and a noise-scaled shape-distance threshold. Defaults are SNR at least eight, at most three mutual neighbors, distance at most 1.5, and weight $e^{-\text{distance}}$; the precise distance is given in Appendix D.1. With the graph held fixed, the gradient is

$$\nabla F = B^\top((BC - Y)/\Sigma^2)/n + \lambda_s R^\top RC + \eta H_w C + \gamma H_w C L_g. \tag{16}$$

The defaults $(\lambda_s, \eta, \gamma) = (0.02, 10^{-5}, 0.05)$ give a strictly convex quadratic because $\eta H_w \succ 0$. Nonnegative L-BFGS-B uses a diagonal Hessian scaling. The exact fixed-grid minimizer is therefore unique. The implementation records a library termination flag and an unscaled projected-gradient residual; it does not compute a certified distance to that minimizer. Termination by a small objective change alone is insufficient to assert accurate first-order stationarity in an ill-conditioned quadratic. Fixed-grid uniqueness also does not establish identifiability of a continuous measure or optimality of the grid search.

Resolution adaptation starts from 128 bins, proposes midpoint splits while conserving parent mass, and allows at most 256 bins. For a candidate split, let $t_h = (k_{h,L} - k_{h,R})/2$. Its score is the sum across spectral channels of squared residual correlations divided by $\|t_h/\Sigma_\ell\|^2$. Splits require sufficient signal information, $\max_\ell \|t_h C_{h\ell}/\Sigma_\ell\|_2/\sqrt{n} \geq 0.25$. Up to a third of current cells are proposed. A refined grid is accepted only with successful inner optimization and a validation MSE below 0.99 of its predecessor; otherwise refinement stops. The column scale is estimated from fitting rows and retained, while the spectral graph is rebuilt using all supplied reconstruction rows for the final fit on the selected grid. This chooses resolution, not the number of molecular species.

4.5. Adaptive Haar RAI: A Convex Problem on Each Tree

The adaptive solver used by RAI-Net is a separate construction. It uses fitting-row scales $s_\ell = \max(\max_{i\in T} |Y_{i\ell}|, 8\,\text{median}_{i\in T}\,\sigma_{i\ell})$, retained for the final refit. Both the data and noise are divided by these scales, and physical masses are restored afterwards. A dyadic tree $\mathcal{T}$ partitions a 256-cell fine log grid into m active leaves, initially eight and at most 64. If leaf j has width w_j, let $U_{j\ell}$ denote its mass divided by $\sqrt{w_j}$. The mass-preserving prolongation and observation matrix are

$$(P_\mathcal{T})_{hj} = \frac{\Delta x_h}{\sqrt{w_j}} \mathbf{1}_{\{h\subset j\}}, \qquad C = P_\mathcal{T} U, \qquad B_\mathcal{T} = B_{\text{fine}} P_\mathcal{T}. \tag{17}$$

Let $Q_\mathcal{T}$ be the orthogonal Haar transform of these normalized leaf coefficients; its first row represents total mass. In normalized data units, joint adaptive RAI solves

$$\min_{U\geq 0} F_{\mathcal{T},\lambda}(U) = \frac{\|(B_\mathcal{T} U - Y)/\Sigma\|_F^2}{2n} + \frac{\eta}{2}\|U\|_F^2 + \lambda\sqrt{p}\sum_{h=2}^{m} \|(Q_\mathcal{T} U)_{h,:}\|_2, \quad \eta = 10^{-4}. \tag{18}$$

The group penalty promotes shared locations of density changes across spectral channels while leaving their amplitudes distinct. The separate-spectral channel variant replaces it by $\lambda \sum_{h\geq 2,\ell} |(Q_\mathcal{T} U)_{h\ell}|$. Neither penalizes the root coefficient. The joint penalty is convex; the ridge makes each fixed-tree problem strictly convex.

For $\lambda > 0$, introduce $V = U \geq 0$ and $Z = QU$. Write $A_\ell = \mathrm{diag}(\Sigma_\ell^{-1})B_\mathcal{T}/\sqrt{n}$, $y'_\ell = Y_\ell/(\sqrt{n}\Sigma_\ell)$, $H_\ell = A_\ell^\top A_\ell + \eta I$, and $c_\ell = A_\ell^\top y'_\ell$. With scaled duals d_V, d_Z and $\rho = 0.5$, the implemented ADMM steps are

$$\begin{aligned}(H_\ell + 2\rho I)U_\ell^+ &= c_\ell + \rho\{V_\ell - d_{V,\ell} + Q^\top(Z_\ell - d_{Z,\ell})\}, \\ V^+ &= [U^+ + d_V]_+, \qquad Z^+ = \mathrm{prox}_{\lambda\sqrt{p}\,\|\cdot\|_{2,1}^{\mathrm{detail}}/\rho}(QU^+ + d_Z), \\ d_V^+ &= d_V + U^+ - V^+, \qquad d_Z^+ = d_Z + QU^+ - Z^+.\end{aligned} \tag{19}$$

The proximal map leaves the first row unchanged and sends a detail row z to $(1 - \lambda\sqrt{p}/(\rho\|z\|_2))_+ z$. For $\lambda = 0$, an augmented ridge NNLS problem is used instead. A feasible primal value and a feasible dual bound give a computable gap; the archived stopping rule is gap $\leq 10^{-5}(1 + F)$, with at most 3000 iterations. Appendix D.2 gives the actual dual construction, so numerical convergence does not depend only on a library success flag.

Tree actions include joint and private residual-detail splits, position-sensitive and occupied-region splits, and sibling merges. Split scores remove numerically visible nuisance tangent directions before correlating residuals with a mass-redistribution contrast. At fixed λ, a split is accepted only if both inner solves meet their gap criterion and

$$F_{\mathrm{old}} - F_{\mathrm{new}} > \max\{10^{-9}(1 + F_{\mathrm{old}}), \mathrm{gap}_{\mathrm{old}} + \mathrm{gap}_{\mathrm{new}}\}. \tag{20}$$

Merges allow a loss within $10^{-10}(1 + F_{\mathrm{old}})$. Visited trees are excluded. The deterministic solver evaluates the available proposals and favors the lowest objective, then fewer leaves. These checks compare attained objectives across representations; they are not a convexity theorem for the whole adaptive search. Completed paths for $\lambda \in \{0, 0.01, 0.1\}$ are compared by noise-standardized internal validation MSE. Among paths within 0.005 of the minimum, the largest λ is chosen, then the selected tree is refitted on all reconstruction rows. Validation selects completed paths; it does not replace Equation (20) for intermediate actions.

4.6. RAI-Net: Learning the Order of Admissible Actions

RAI-Net retains Equations (18)–(20). For a tree state and a proposed action a, it predicts a scalar priority

$$q_\theta(\phi_a) = w_3^\top \tanh\big(W_2 \tanh(W_1\widetilde{\phi}_a + b_1) + b_2\big) + b_3, \qquad \widetilde{\phi}_a = (\phi_a - \mu)/s. \tag{21}$$

The 14 features encode current and proposed tree sizes, action type, conditional and raw residual scores, information, occupied mass, affected width, penalty, data dimensions, and the current objective; Appendix D.3 specifies them. Feature standardization is fitted on training records only, with each scale at least 0.1. No true diffusion rates or species count are inference inputs.

For recorded action gain g_a, let $t_a = \mathrm{sign}(g_a)\log(1 + |g_a|)$. Training minimizes

$$\frac{1}{N}\sum_{a=1}^{N} \ell(q_\theta(\phi_a) - t_a), \qquad \ell(e) = \begin{cases} e^2/2, & |e| < 1, \\ |e| - 1/2, & |e| \geq 1. \end{cases} \tag{22}$$

The original policy's target is training-objective decrease. In the retrained policy, it is noise-standardized internal validation prediction-MSE decrease; failed actions receive $g_a = -1$. The latter uses two 48-unit hidden layers, full-batch Adam with learning rate 0.003, 400 epochs, and seed 9,252,026. Whole-problem training and validation partitions, rather than random action-row splits, avoid sharing a problem across these partitions. The bank

and checkpoint selection are documented in Section 6. Changing the target and the training bank together means this is a policy comparison, not an isolated architecture ablation.

At inference, actions are sorted by decreasing q_θ. With the archived shortlist of one, evaluation continues until a numerically admissible action is available; failed proposals lead to further candidates. Invalid policy output falls back to the deterministic order. The accepted action must still satisfy the classical gap and descent checks. The network therefore neither returns a density directly nor classifies the number of components. Its ordering can change the finite search path because not every action is evaluated: numerical safeguards are retained, but equality with deterministic RAI's final answer is not guaranteed.

4.7. DOME: Atomic Inversion with Moment-Based Proposals

DOME uses a common known σ and solves the corresponding fixed-order atomic residual problem in Equation (11); its difference lies in proposal generation, nonlinear refinement, and native model selection. It normalizes the whole matrix and noise by $\max(\max|Y|, \sigma)$. Internally it uses $D_{\mathrm{ref}} = 10^{-9}\,\mathrm{m^2\,s^{-1}}$, $t_i = b_i D_{\mathrm{ref}}$, $\delta_j = d_j/D_{\mathrm{ref}}$, and $u_j = \log(\delta_j/\delta_{\min})/\log(\delta_{\max}/\delta_{\min}) \in [0,1]$. Thus

$$\partial_{u_j} k_j = -t\,\delta_j \log(\delta_{\max}/\delta_{\min}) \odot k_j. \tag{23}$$

At fixed rates, the small NNLS problem is solved by enumerating all active subsets, including the empty subset, using SVD least squares. A bounded trust-region reflective solve then uses the full active-set residual Jacobian of Proposition 6. This is distinct from native atomic RAI's L-BFGS-B envelope-gradient search and from original TRAIn's squared-amplitude iteration.

The following proposal equations use these dimensionless rates and acquisition coordinates. A split of mass a at rate m into πa at $m - (1-\pi)\Delta$ and $(1-\pi)a$ at $m + \pi\Delta$ preserves mass and first moment. At fixed t,

$$Y_{\mathrm{split}}(t) - ae^{-tm} = \tfrac{1}{2} ae^{-tm} t^2 \pi(1-\pi)\Delta^2 + O(\Delta^3). \tag{24}$$

Let $E = KA - Y$, $P_K = KK^+$, and J denote the profiled residual Jacobian in the internal normalized units. With consistent rate units, a curvature proposal for component j uses

$$h_j = (I - P_K)(t^2 \odot k_j/2), \quad q_j = (I - JJ^+)\,\mathrm{vec}(h_j A_{j,:}), \quad \widehat{v}_j = \left[-\frac{\langle \mathrm{vec}\,E, q_j\rangle}{\|q_j\|_2^2}\right]_+. \tag{25}$$

Zero or numerically negligible denominators provide no informative direction. A complementary spectral contrast uses

$$v_j = (I - P_K)(-t \odot k_j), \qquad \zeta_\ell = -\tanh\!\left(\frac{v_j^\top E_\ell}{\max(\sigma\|v_j\|_2, 10^{-15})}\right), \qquad T_{i\ell} = v_{j,i} A_{j\ell} \zeta_\ell. \tag{26}$$

Its candidate half-separation is $\widehat{s}_j = [-\langle E, T\rangle / \|T\|_F^2]_+$. The implementation proposes standard deviation scales $s \in \{\sqrt{\widehat{v}_j}, \widehat{s}_j, 0.12\delta_j, 0.4\delta_j, 0.85\delta_j\}$ and fractions $\pi \in \{0.15, 0.5, 0.85\}$, setting $\Delta = s/\sqrt{\pi(1-\pi)}$ before restricting the pair to the physical bounds. Every proposal is evaluated with exact exponentials, not with the truncated expansion.

Pairwise exchanges use aggregate masses $M_j = \sum_\ell A_{j\ell}$. Given a pair's mass-normalized mean m and variance v, alternative fractions $\pi \in \{0.08, 0.2, 0.5, 0.8, 0.92\}$ yield

$$\delta_L = m - \sqrt{v(1-\pi)/\pi}, \qquad \delta_R = m + \sqrt{v\pi/(1-\pi)}. \tag{27}$$

This preserves the pair's aggregate first two moments at proposal construction. It does not preserve each spectral channel's moments, and subsequent NNLS refinement is not constrained to preserve them. A finite beam, distinct-seed filtering, and local refinement limit the search. Physical rates are returned as $d_j = D_{\mathrm{ref}}\delta_j$; the internal numerical floors apply before this conversion. The native selector includes the zero model and minimizes

$$\mathrm{score}_{\mathrm{DOME}} = \mathrm{RSS}/\sigma^2 + 2d_f + \frac{2d_f(d_f+1)}{\max(np - d_f - 1, 1)}, \quad d_f = r + \#\{A_{j\ell} > 10^{-10}\}, \tag{28}$$

where the amplitude cutoff is in DOME's normalized units. This is a heuristic effective-parameter count, not a regular-mixture likelihood theorem. Native status recording is distinct from a convergence-gated selector. DOME-S replaces the final score by the common procedure in Section 5; the present study contains no ablation proving an independent advantage of moment exchange.

4.8. TRAIn-MF: A Shared Distributional Factorization

TRAIn-MF extends the single-channel TRAIn method of Xu and Zhang [4] to simultaneous inversion across spectral channels [19]. In $Y \approx KSA$, the columns of S describe shared diffusion profiles and the rows of A their spectral amplitudes. The model remains multifrequency at rank one, and a profile may be broad or multimodal.

For nonzero data, let $s_Y = \|Y\|_F/\sqrt{np}$ and $\widetilde{Y} = Y/s_Y$; use $s_Y = 1$ for zero data. The regularized model solves

$$\min_{S\geq 0,\, \mathbf{1}^\top S = \mathbf{1}^\top,\, A\geq 0} F(S,A) = \frac{\|KSA - \widetilde{Y}\|_F^2}{2p} + \frac{\lambda_S}{2}\|LS\|_F^2 + \frac{\lambda_A}{2p}\|A\|_F^2. \tag{29}$$

It uses at least 256 mass bins. With log nodes x_h, quadrature weights w_h, and $\rho_h = S_h/w_h$, L discretizes density curvature:

$$(LS)_h = \frac{1}{\sqrt{\overline{\Delta}_h}}\left\{\frac{\rho_{h+1}-\rho_h}{\Delta_h} - \frac{\rho_h - \rho_{h-1}}{\Delta_{h-1}}\right\}, \quad \overline{\Delta}_h = (\Delta_{h-1} + \Delta_h)/2, \tag{30}$$

for interior nodes, with trapezoidal w_h. This avoids treating grid masses as density heights and imposes no extra endpoint curvature rows. The automatic reference sets $\lambda_S = \lambda_A = 10^{-8}$. Global RMS normalization makes these penalties relative to the observed signal scale. Returned amplitudes are multiplied by s_Y.

For fixed S, each amplitude column is obtained by the Lawson–Hanson active-set NNLS algorithm [22], applied to the ridge-augmented system:

$$A_\ell^+ = \arg\min_{a\geq 0} \left\| \begin{bmatrix} KS \\ \sqrt{\lambda_A} I \end{bmatrix} a - \begin{bmatrix} \widetilde{Y}_\ell \\ 0 \end{bmatrix} \right\|_2^2. \tag{31}$$

The positive ridge makes this conditional solution unique. The curvature penalty alone does not guarantee uniqueness of the profile update; Proposition A1 supplies a sufficient condition and a degeneracy counterexample. The frozen runtime checks feasibility and KKT residuals but does not test that injectivity condition. No uniqueness claim is inferred from a successful KKT flag. Appendix E gives the distinction and a separate executable numerical diagnostic; this post hoc diagnostic is not represented as a check that ran in the archived benchmark. For fixed A, the S update is a joint simplex-constrained quadratic, not

a collection of independently clipped residual fits. If $A^\top = Q_A R_A$ is a thin QR factorization, orthogonality gives

$$\|KSA - \widetilde{Y}\|_F^2 = \|KSR_A^\top - \widetilde{Y}Q_A\|_F^2 + \|\widetilde{Y}(I - Q_A Q_A^\top)\|_F^2. \tag{32}$$

Consequently, with columnwise vectorization $s = \operatorname{vec} S$, this update solves

$$\min_{s\geq 0,\,(I_r\otimes \mathbf{1}^\top)s=\mathbf{1}} \frac{1}{2}\left\| \begin{bmatrix} (R_A \otimes K)/\sqrt{p} \\ \sqrt{\lambda_S}(I_r \otimes L) \end{bmatrix} s - \begin{bmatrix} \operatorname{vec}(\widetilde{Y}Q_A)/\sqrt{p} \\ 0 \end{bmatrix} \right\|_2^2. \tag{33}$$

An active-set algorithm for linearly constrained least squares supplies a feasible proposal, with an interior-point fallback. Along its feasible segment from the current point, an exact scalar quadratic line search clips the step to $[0, 1]$ and checks the attained objective.

The implementation uses at most eight alternating sweeps as a warm start, not as its convergence criterion or total iteration budget. When the KKT test has not passed, it then refines the profiled objective $\Psi(S) = \min_{A\geq 0} F(S, A)$ by simplex-constrained sequential quadratic programming (SQP), recomputing Equation (31) at every evaluation. Its envelope gradient is

$$G_S = K^\top(KSA - \widetilde{Y})A^\top/p + \lambda_S L^\top LS. \tag{34}$$

For a simplex column s_j, set $\nu_j = s_j^\top G_{S,j}$. Stationarity requires $G_{S,j} - \nu_j \mathbf{1} \geq 0$ and $s_j \odot (G_{S,j} - \nu_j \mathbf{1}) = 0$, together with amplitude KKT conditions for $G_A = (KS)^\top(KSA - \widetilde{Y}) + \lambda_A A$. The code checks scaled primal, dual, and complementary residuals at tolerance 10^{-6}, uses two starts, and records unresolved optimization. Accepted objective values are nonincreasing within numerical tolerance and bounded below; this alone does not prove convergence of the factors, uniqueness, or a global minimum. General block-coordinate results require hypotheses beyond observing descent [49]. In the two archived starts underlying the displayed three-profile example, the scaled KKT residuals after eight sweeps were 6.77×10^{-3} and 2.83×10^{-3}. Subsequent SQP refinement used 250 and 235 iterations, respectively, and reduced them to 5.56×10^{-8} and 7.01×10^{-9}. The first start reached the SQP iteration limit (exit flag zero), although its final independently evaluated KKT residual passed the declared tolerance; the selected second start used 235 iterations. This audit demonstrates that eight sweeps alone were insufficient in that example. The separate controlled study in Appendix G.1 compares four warm-start budgets across ranks and SNRs with identical subsequent SQP budgets. It reports all unresolved fits rather than assuming adequacy or optimality of eight.

In the controlled convergence study (Appendix G.1), the curvature-null-space condition is checked before each S block and at each reported SQP iterate, including iteration zero. For the interior operator in Equation (30) with $\lambda_S > 0$, the check uses the reduction in Appendix E. An unresolved condition routes the profile block to the interior-point solver with the same objective and constraints. This conditional check does not establish convergence of the full factorization or identifiability of the factors. The check was not run at each iterate of the fit shown in Figure 4.

4.8.1. Automatic Factor Selection in TRAIn-MF

This selector is distinct from the RAI-S/DOME-S rule. For n_V internal validation rows and known σ, define

$$\tau = \sigma\{\sqrt{n_V} + \sqrt{p} + \sqrt{2\log(1/0.01)}\}, \qquad r_{\text{res}} = \#\{s_j(Y_V) > \tau\}. \tag{35}$$

Candidates $r = 1, \ldots, \min(4, r_{\text{res}})$ are fitted on internal fitting rows. If no direction is resolved, the estimate is zero. Among numerically converged candidates, let r_* minimize the mean validation row loss $\ell_{i,r} = \|\widehat{Y}_{i,r} - Y_i\|_2^2/p$. Put $\delta_{i,r} = \ell_{i,r} - \ell_{i,r_*}$. Selection takes the smallest rank satisfying

$$\overline{\delta}_r \le \mathrm{sd}_{i \in V}(\delta_{i,r})/\sqrt{n_V} + 10^{-15}. \tag{36}$$

This is an empirical paired standard error across acquisitions, not the conditional Gaussian prediction-distance allowance in Equation (44). Failed candidate optimization, a failed final all-row refit, or $r_{\text{res}} > 4$ generates an unresolved status. Since rank counts visible signal directions, its use as a search restriction can miss chemical species with proportional spectra (Proposition 4); r_{res} is not an upper bound on their number. Selection and signal masking also preclude interpreting Equation (35) as an unconditional chemical detection guarantee.

5. Nonnegative Variable Projection and Automatic Support Selection

5.1. Conditional Amplitudes and the Exact Local Derivative

For r candidate rates, introduce dimensionless coordinates $z_j = \log(D_j/D_0)$ and a binary allowed-support matrix $H \in \{0,1\}^{r \times p}$. Conditional amplitudes solve

$$A_H(z) \in \underset{A \ge 0,\, A_{j\ell}=0 \text{ if } H_{j\ell}=0}{\arg\min} \tfrac{1}{2}\|\sigma^{-1}(K(z)A - Y)\|_F^2, \quad \psi_H(z) = \tfrac{1}{2}\|R_H(z)\|_F^2. \tag{37}$$

Here $R_H = (KA_H - Y)/\sigma$. Rates are bounded by $[D_{\min}, D_{\max}]$. Only selected signal spectral channels enter this matrix problem; every retained spectral channel is fitted jointly. The method neither selects a single representative peak nor fits all 65,536 points of an otherwise sparse experimental spectrum indiscriminately.

With fixed rates and support, columns separate. If each allowed kernel submatrix has full column rank, the corresponding NNLS minimizer is unique. Without that condition, the fitted vector remains unique as the Euclidean projection onto a closed finitely generated cone, although the amplitudes can be nonunique. In exact arithmetic, enumerating all active subsets and selecting a feasible least-squares minimum solves the small NNLS problem. The implementation uses this strategy with SVD-based linear solves; it has exponential cost in r, so it is appropriate to the present cap of four rates, not to a dense 256-variable NNLS problem. The TRAIn-MF implementation instead uses the Lawson–Hanson active-set NNLS algorithm [22].

For allowed amplitudes, the Karush–Kuhn–Tucker conditions are

$$A \ge 0, \qquad G_A = K^\top(KA - Y)/\sigma^2 \ge 0, \qquad A \odot G_A = 0. \tag{38}$$

No sign condition on G_A is required for coordinates fixed to zero by H: their feasible directions are absent. This distinction matters when auditing a screened solution.

Proposition 6 (Residual derivative on a stable active set)**.** *Consider one column with locally unchanged positive active set J, full-column-rank K_J, and strict inactive inequalities. Set $a = K_J^+ y$, $r = (K_J a - y)/\sigma$, and $P_J = K_J K_J^+$. For an infinitesimal change dK_J,*

$$dr = (I - P_J)(dK_J)a/\sigma - K_J^{+\top}(dK_J)^\top r. \tag{39}$$

Consequently, $d(\frac{1}{2}\|r\|^2) = r^\top(dK_J)a/\sigma$.

Proof. The active normal equations give $K_J^\top r = 0$. Differentiating this identity and $\sigma r = K_J a - y$ yields $(K_J^\top K_J)da = -(dK_J)^\top \sigma r - K_J^\top (dK_J)a$. Substitution into dr gives Equation (39). Multiplication by $r^\top$, using $r^\top K_J = 0$, gives the reduced differential. □

For a log-rate perturbation, $\partial K_{ij}/\partial z_j = -b_i D_j e^{-b_i D_j}$. The residual-dependent second term in Equation (39) is retained. Dropping it changes the residual Jacobian even though the reduced objective gradient still has the envelope form. At active-set changes or rate collisions this smooth formula is not a differentiability theorem. A bounded trust-region reflective solver recomputes NNLS at every evaluation. In the archived code, $\log D_j$ is used directly; it differs from z_j by a constant and has the same derivative.

Termination is checked through the library status, the amplitude KKT residual, and a scaled projected log-rate gradient. Specifically, with $g = J^\top \operatorname{vec} R$ and log-bound box $\mathcal{B}$, the recorded rate residual is

$$\eta_z = \frac{\|z - \Pi_{\mathcal{B}}(z - g)\|_\infty}{\max(1, \|J\|_2 \|R\|_F)}. \tag{40}$$

The denominator makes this a numerical diagnostic with an explicit scale, not a statistical precision guarantee. We require $\eta_z \le 10^{-6}$ and a relative amplitude KKT residual at most 10^{-7} for candidate eligibility. A failed candidate is recorded. These checks certify approximate first-order behavior of the attained candidate, not global minimization of the nonlinear problem or exhaustive support search.

5.2. Spectral Coherence Through Residualized Kernels

Let the retained spectral channels be partitioned into contiguous groups $G_1, \dots, G_g$. For a candidate rate D_j, remove its column from K, writing the remaining matrix K_{-j}. Define

$$v_j = (I - K_{-j} K_{-j}^+) k_j, \qquad Z_{jG} = \frac{\sum_{\ell \in G} v_j^\top y_\ell}{\sigma \|v_j\|_2 \sqrt{|G|}}. \tag{41}$$

The projection removes the linear span of the other candidate decays. It need not identify the true nuisance components. For fixed correct rates, it provides a useful elementary calculation.

Proposition 7 (Fixed-design screening identity)**.** *Suppose the rates and the spectral-channel groups are fixed independently of iid Gaussian errors, the data follow $Y = KA + E$, and $v_j \neq 0$. Then*

$$Z_{jG} \sim N\left(\frac{\|v_j\|_2 \sum_{\ell \in G} A_{j\ell}}{\sigma \sqrt{|G|}}, 1 \right). \tag{42}$$

If component j is absent throughout G, its mean is zero. Testing rg such fixed nulls at threshold $\Phi^{-1}(1 - \alpha/(rg))$ controls the probability of at least one false rejection by α.

Proof. The nuisance term vanishes because $v_j^\top K_{-j} = 0$, while $v_j^\top k_j = \|v_j\|_2^2$. Summing independent normal errors gives variance $|G|\sigma^2 \|v_j\|_2^2$ before normalization. The final claim is the union bound and does not require independence between the different tests. □

When amplitudes are equal to a in a group of m spectral channels, the mean score is $a\|v_j\|_2 \sqrt{m}/\sigma$. Coherent small amplitudes can therefore be retained even if each is small relative to an unrelated tall peak. Adding spectral channels with no component signal instead dilutes this score. As kernels become collinear, $\|v_j\|_2$ decreases, expressing the loss of separation information (Figure 1). With correlated spectral errors, the variance generally

changes; replacing it by $m\sigma^2$ is not justified. Proposition A2 gives the fixed-covariance form and identifies the assumptions needed for such an extension.

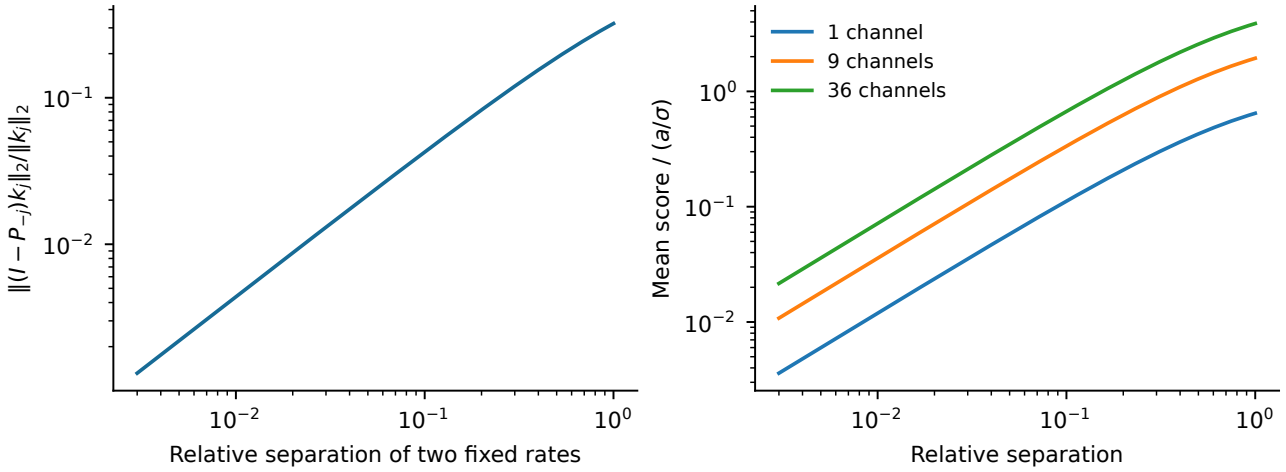


Figure 1. Fixed-design illustration of Equation (42), not a reconstruction benchmark. The two rates are $D/D_0 = 1$ and $1 + \Delta$, with 32 equally spaced bD_0 values from 0 to 2.2. Left: the residualized-kernel norm decreases near collision. Right: coherent independent spectral channels increase the expected standardized score by $\sqrt{m}$ for equal amplitudes. The rates and groups are fixed in this calculation.

The operational screen estimates rates from the fitting data and groups spectral channels after a signal mask. Consequently, Proposition 7's calibration assumptions do not apply to the full estimator. Independent masking alone would not remove all uncertainty: an estimated nuisance kernel can leave a nonzero residualized mean even when the tested component is absent. The fixed-design result requires correct nuisance rates, or a separately justified bound on this model error. The additional null experiments in Appendix F assess the operational procedure under their specified generators; they do not replace these assumptions. We use $\alpha = 0.01$ as a frozen screening parameter, not as a claimed family-wise error rate or chemical confidence level. If $\|v_j\|_2 < 10^{-10}\|k_j\|_2$, the implementation retains that component's allowed support: numerical nonidentifiability is not evidence of absence. Otherwise an entire group is allowed only when Z_{jG} exceeds the threshold. NNLS still estimates separate nonnegative amplitudes at every retained spectral channel within an allowed group.

5.3. Predictive Comparison and Automatic Order

The available reconstruction rows are divided into an internal fitting set T and an internal validation set V. Candidate paths include orders $0, 1, \ldots, 4$. For each path state, both the unrestricted support and, when different, the screened support are refined on T. A component with no allowed spectral channel is removed before refinement. Write $P_c = \widehat{Y}_{c,V} = K(b_V, d_c)A_c$ for a candidate's prediction on V. Both P_c and P_{c_*} below are predictions; neither is an observed-data residual. Put

$$L_c = \|P_c - Y_V\|_F^2/\sigma^2. \tag{43}$$

For numerically eligible candidates, let c_* minimize L_c. We retain the set

$$\mathcal{E} = \left\{c : L_c \le L_{c_*} + 2\|P_c - P_{c_*}\|_F/\sigma + 10^{-10}\right\}. \tag{44}$$

Within $\mathcal{E}$, selection is lexicographic: fewer surviving rates, then fewer allowed amplitudes, then lower validation loss. The selected fixed support is refitted on $T \cup V$. The zero-signal model is a legitimate candidate. The selected r is therefore automatic within a

declared finite search, without access to the true order. The selected order and configured cap are reported. A selection at that boundary does not establish that no additional rates exist.

Proposition 8 (Conditional paired prediction variance)**.** *Let P, Q be fixed predictions independent of E, and $Y = M + E$ with iid $N(0, \sigma^2)$ errors. Then*

$$\mathbb{E}\{(\|P - Y\|_F^2 - \|Q - Y\|_F^2)/\sigma^2\} = (\|P - M\|_F^2 - \|Q - M\|_F^2)/\sigma^2, \tag{45}$$

$$\mathrm{Var}\{(\|P - Y\|_F^2 - \|Q - Y\|_F^2)/\sigma^2\} = 4\|P - Q\|_F^2/\sigma^2. \tag{46}$$

Proof. Subtracting the losses cancels the quadratic error term. The random remainder is $-2\langle P - Q, E\rangle/\sigma^2$, a centered normal variable with the variance in Equation (46). □

For a fixed pair, define $s(c, d) = 2\|P_c - P_d\|_F/\sigma$. Proposition 8 gives exactly this standard deviation, rather than a residual-norm approximation, when the pair is independent of validation noise. Equation (44) uses $s(c, c_*)$ as a predictive simplicity allowance. Because c_* is chosen using the validation losses, the fixed-pair identity does not calibrate this adaptive decision or supply a simultaneous confidence set. In particular, the one-standard deviation multiplier is a declared heuristic; it is not a significance level or a probability that a component is real. Moreover, the benchmark signal mask uses all outer training rows, some of which become internal validation rows. Conditional independence of predictions and internal validation noise is therefore not exact after masking. The internal rule is explicitly heuristic. The eight external test acquisitions remain unused in the mask, candidate fitting, and selection, preserving an independent predictive evaluation for this full procedure. A future inferential version would need independent masking or cross-fitting and selection-adjusted calibration; neither has been retroactively attributed to the frozen benchmark.

The suffix S specifies the complete estimator below: the objective for a fixed (r, H) is Equation (37), the proposal path is native atomic RAI or DOME (Sections 4.3 and 4.7), and the final order/support rule is Equation (44). Screening defines a finite family of constraints; it is not an extra amplitude penalty. In particular, rates are screened at the native path state before each support-specific refinement. The screen is not iterated after every rate update. Algorithm 1 records this order of operations and includes explicit failure handling.

Algorithm 1 Automatic spectral-support selection for RAI-S and DOME-S.

Require: Observed (b, Y, σ), retained spectral channels, diffusion bounds, path method M, $r_{\max} = 4$.
Ensure: Rates d, amplitudes A, allowed support H, selected order, diagnostics, and status.
Sort distinct b; partition supplied rows into fitting T and validation V.
Form contiguous spectral-channel groups without crossing large chemical shift gaps.
$\mathcal{P} \leftarrow \textsc{NativePath}(M, b_T, Y_T, \sigma, r_{\max})$; include the zero model.
$\mathcal{C} \leftarrow \varnothing$; $\mathcal{L} \leftarrow \varnothing$ ▷ Eligible candidates and audit log
for each rate state $d \in \mathcal{P}$ **do**
  $\mathcal{H} \leftarrow \{\mathbf{1}_{|d|\times p}\}$.
  if $|d| > 0$ **then**
    $H_s \leftarrow \textsc{Screen}(b_T, Y_T, \sigma, d, \text{groups}, 0.01)$ using Equation (41).
    Add H_s to $\mathcal{H}$ if different from the unrestricted support.
  end if
  for each $H \in \mathcal{H}$ **do**
    $(d_H, H) \leftarrow \textsc{PruneEmptyRows}(d, H)$.
    $c \leftarrow \textsc{RefineNNLS}(b_T, Y_T, \sigma, d_H, H)$ using Equations (37)–(40).
    Append rates, support type, parameter count, success and KKT to $\mathcal{L}$.
    if c is the zero model, or solver success, $\eta_z \le 10^{-6}$ and KKT $\le 10^{-7}$ **then**
      $P_c \leftarrow K(b_V, d_c)A_c$; $L_c \leftarrow \|P_c - Y_V\|_F^2/\sigma^2$.
      Add (c, P_c, L_c) to $\mathcal{C}$.
    end if
  end for
end for
if $\mathcal{C} = \varnothing$ **then**
  return unresolved optimization and $\mathcal{L}$.
end if
$c_* \leftarrow \arg\min_{c\in\mathcal{C}} L_c$; construct $\mathcal{E}$ by Equation (44).
$\widehat{c} \leftarrow \arg\min^{\text{lex}}_{c\in\mathcal{E}}(|d_c|,\ |d_c| + \#\{H_c = 1\},\ L_c)$.
Refit $(d_{\widehat{c}}, A_{\widehat{c}})$ on $T \cup V$ with $H_{\widehat{c}}$ fixed; check diagnostics.
Export a mass-conserving 256-bin map; retain final failure or mismatch flags.
return final (d, A, H), selected order, status, candidate losses, and $\mathcal{L}$.

6. Reproducible Computational Study

6.1. Data Partitions and Prespecified Decisions

The primary study contains 24 development problems and 48 new test problems, generated with disjoint seed ranges beginning at 510,000 and 810,000. A separate training bank for a neural policy comparison begins at 610,000. A problem is an entire noisy DOSY matrix, not an individual spectral pixel. The support configuration was frozen before test evaluation: maximum order four, screening parameter 0.01, validation stride four, one paired standard deviation allowance, and at most 350 nonlinear function evaluations per support refinement.

Support selection gives priority to smaller component order, then to fewer allowed amplitudes (Equation (44)). The screening parameter was not tuned on test data. Appendices F and G report additional tests of the prespecified procedure.

Each new problem has 32 acquisition coordinates and 1024 spectral channels uniformly spanning 10 to 0 ppm. In the primary atomic benchmark, $D_0 = 10^{-9}\,\mathrm{m^2\,s^{-1}}$ and fitted rates are restricted to $[0.1, 15]D_0$. The native rate constraints in the external comparisons are specified in Appendix G.5. The acquisition design is

$$b_i = 10^9 B\left(\frac{i}{31}\right)^{\eta}\ \mathrm{s\,m^{-2}}, \quad i = 0, \ldots, 31, \tag{47}$$

with independent $B \sim U(1.3, 2.7)$ and $\eta \sim U(1.5, 2.4)$. The component diffusion coefficients are positive and jointly shared across all spectral channels. Their amplitudes have correlated Gaussian spectral peaks, including an overlapping region near 7.73 ppm and other peaks with different mixture proportions. The exact generator is included with the evidence package.

Eight families contain six problems each: one, two, or three separated rates; a three-rate weak-component family; a close-rate family; proportional component spectra; noise-only controls; and broad distributions. The six problems per family use SNR 50, 100, and 200 twice each, with independently generated parameters and noise. For positive signals, the maximum unattenuated spectral intensity is normalized to one in the generator and $\sigma = 1/\mathrm{SNR}$. Solvers receive observations in these units, not true component normalizations.

Nominal separated rates are $(0.65, 2.3, 7.1)D_0$, using the first one or two as appropriate, with independent log-rate perturbations of standard deviation 0.10. Close rates use $(0.8, 1.05, 5.5)D_0$ with log-rate perturbations of standard deviation 0.02. In the weak family the third spectrum is multiplied by 0.04 before overall normalization; this is a spectral-amplitude multiplier, not a guarantee of exactly 4% total mixture mass. In the proportional family $A_{j\cdot}$ equals $1, 0.7, 0.45$ times the same spectrum. Broad controls replace each atom by a lognormal distribution with log-width 0.18, using 24-point Gauss–Hermite forward quadrature. The forward generator uses the lognormal on its full positive domain; its histogram reference is truncated and renormalized on the fitted interval for plotting and shape evaluation. These controls deliberately violate the fitted atomic model.

Every fourth acquisition, using zero-based indices $i \bmod 4 = 3$, is reserved for external testing. The remaining 24 rows are available to the reconstruction. A signal mask thresholds the mean of the first six available low-b rows at $4\sigma/\sqrt{6}$, followed by two one-dimensional dilation steps. Noise-only controls instead inspect the fixed window $4 < \mathrm{ppm} < 4.65$. All methods in a comparison receive the same mask. Conditional on that mask, external test noise remains independent. The reported errors assess retained spectral channels, not undetected parts of a full spectrum; losses caused by masking are not chemical absence certificates.

After the 24 reconstruction rows are sorted by increasing acquisition coordinate, the 3rd, 7th, 11th, 15th, 19th and 23rd rows form the internal validation set. The other 18 rows are used for candidate fitting. These are one-based positions within the reconstruction subset, not positions in the original 32-row acquisition. Groups are split where adjacent selected chemical shift spacings exceed 1.6 times their 25th percentile. This rule assumes a substantially regular original spectral grid. It is a segmentation convention, not a learned chemical grouping. The final selected model uses all 24 reconstruction rows.

6.2. Evaluation, Denominators, and Uncertainty Summaries

The primary quantitative panel is the 36 signal-present atomic problems. Six null cases are reported separately because normalized shape and true spectral leakage have no physical denominator there. Six broad cases are misspecification controls. An additional 24-problem archived bank contains finite-width diffusion profiles with one to three generating components, including close and proportional scenarios. It is transfer evidence and is never pooled with the new atomic bank.

Rates are paired to ground truth by a minimum-cost one-to-one assignment in absolute log distance; pairs beyond 0.25 are then rejected. “Order and rates recovered” requires the correct total order and an accepted pair for every true rate. The tolerance corresponds to an approximate multiplicative factor of $e^{0.25}$ and is not a chemical identification standard. On finite-width controls, matching an atomic rate to a distribution center does not imply recovery of the distribution or a literal atom.

For each accepted pair (j,k), define an evaluator-only meaningful spectral set $\mathcal{S}_k = \{\ell : A^{\text{true}}_{k\ell} > 0.001 \max_h A^{\text{true}}_{kh}\}$ within the supplied mask. The leakage fraction is

$$\ell_{\text{spec}} = \frac{\sum_{(j,k)\text{ paired}} \sum_{\ell \notin \mathcal{S}_k} \widehat{A}_{j\ell}}{\sum_{(j,k)\text{ paired}} \sum_{\ell} \widehat{A}_{j\ell}}. \tag{48}$$

The implementation assigns zero when its denominator is zero; therefore unmatched estimated mass is always reported alongside it, as a fraction of all estimated mass. Missing true rates are exposed by the recovery count. The 0.1% threshold is an evaluation convention and never a fitting cutoff. It is also distinct from the display threshold in the DOSY figures.

External clean prediction error is $\|\widehat{Y}_U - Y^0_U\|_F/(\sigma\sqrt{|U|p})$ on the eight unused acquisitions U. It measures estimation error without adding a fresh noise variance; observed noisy-test errors are retained in the machine-readable results. Shape loss is W_1 in natural-log diffusion, evaluated after exporting both measures to the common 256-cell mass grid, normalizing per positive spectral channel, and weighting spectral channels by true mass only in the evaluator. A missing positive estimate receives the declared span penalty log 150; it is not described as an actual transport distance. Total estimated mass is saved separately.

We report problem-level averages with equal weight per problem. The primary table excludes the null cases; Appendix B retains the earlier 42-case convention, whose zero contributions dilute shape averages. Bootstrap intervals resample whole paired problems 10,000 times with seed 9925. They are descriptive pointwise intervals for this small heterogeneous simulation bank, not simultaneous tests or calibrated population coverage. Different spectral channels are not treated as independent experimental replicates.

6.3. Neural Policy Comparison and Numerical Verification

RAI-Net is the action-ranking policy of Section 4.6, acting on the adaptive distributional objective in Equation (18). It does not directly predict the number of species. Retraining used 80 DOSY problems split by whole problem into 64 training and 16 validation problems, producing 2921 recorded action examples. The policy has two hidden layers of 48 units with hyperbolic tangent activations. Training lasted 400 epochs and used a transformed noise-standardized internal validation prediction-gain target. The selected checkpoint was epoch 400, so convergence of learning is not claimed. The classical objective and numerical acceptance checks remained in force.

The original and retrained policies were evaluated on the predefined first 24 new test problems and all 24 archived transfer problems. The new neural panel contains three problems per scenario: 18 signal-present atomic, three null, and three broad distribution cases. These are smaller panels than the atomic study. The network results are an auxiliary comparison, not an additional order recovery experiment or evidence for the atomic support rule.

Numerical verification covers NNLS agreement with an independent implementation, finite-difference checks of the variable-projection Jacobian, coherent weak-signal screening, acquisition partition boundaries, and amplitude-scaling/spectral channel-permutation behavior. A direct comparison of the SVD-based active-subset enumerator with an independent Lawson–Hanson active-set NNLS implementation (MATLAB R2026a Update 2) used 100 amplitude columns. Pseudoinverse-based local rate precision, boundary hits, and a residual-based atomic model warning are exported as diagnostics; none is interpreted as a selection-adjusted confidence interval or goodness-of-fit test.

6.4. Validation Studies and Experimental Sampling Units

Table 3 identifies the independent inputs and repeated conditions in each additional study. Methods are compared on the same observations; additional policies, candidate or-

ders and warm-start budgets do not increase the number of independent samples. The supplement contains the configurations, seeds, checkpoints and failed outcomes. In the convergence study, we fix the factor count to examine optimization alone. The weak-component and experimental studies retain automatic atomic order selection.

Table 3. Sampling units of the revision studies. Repeated conditions and methods do not multiply the number of independent inputs.

Panel	Independent Input and Repeated Conditions
Null	300 noise matrices; 100 per noise scale; paired methods.
Weak	96 truth/noise matrices; four amplitude factors, three SNRs, eight per cell.
Factorial	36 existing atomic problems; paired 2×2 stages and production reference.
TRAIn-MF	Nine rank–SNR inputs; four warm budgets and two common starts.
Neural	120 new test problems; five trained seeds and one classical control on each.
External	47 frozen matrices after one adapter pilot; two stresses on 12 underlying truths.
Experimental	Four resin specimen series with three diffusion-time blocks each.
Public mixture	One 30-acquisition archived experiment; five paired estimators.

We studied four industrial resin series: TP194-2623086, TP194-2623088, TP194-2623084 and TP194-2619082. We retain the signed absorption spectra and fit every channel within the positive support detected from the fitting acquisitions. Each 23-acquisition block reserves five acquisitions for prediction; neither these nor the internal validation rows enter support detection. All methods use the same encoding and support. Coherent negative structure outside the support is reported and left unestimated. Strong correlation in the processed background means that its estimated noise scale cannot be interpreted as an iid Gaussian calibration. We retain the HDO-derived encoding calibration, but its unquantified systematic uncertainty prevents assigning a chemical confidence interval. Appendix G.6 describes the samples and the limits of the available composition records.

7. Results

7.1. Atomic Recovery and Spectral Leakage

Table 4 reports the 36 signal-present atomic test problems. RAI-S reduces mean leakage from 5.166% to 1.728%, and DOME-S from 3.857% to 1.728%. The clean external prediction error decreases from 0.2870 to 0.2330 for RAI and from 0.2591 to 0.2330 for DOME. Binned shape loss decreases from 0.1523 to 0.1094 and from 0.1393 to 0.1094, respectively. These are paired changes on the same observations and masks.

The favorable averages do not hold for every endpoint. Unmatched estimated mass increases from 6.774% to 7.257% for RAI and from 7.019% to 7.257% for DOME. RAI's correct-order count rises from 17/36 to 25/36, but DOME's falls from 26/36 to 25/36. The stricter order-and-rates count improves from 17/36 to 23/36 and from 22/36 to 23/36. Thus, the strongest common conclusion is improved spectral assignment and prediction, not uniformly improved component counting.

Table 4. New atomic test: 36 signal-present problems, excluding null and broad distribution controls. Leakage and unmatched mass are percentages. E_U is clean external RMSE divided by σ; $W_{1,h}$ is the binned log-diffusion shape score. A correct count is weaker than recovery of both order and rates.

Method	Correct Order	Order and Rates	Leakage	Unmatched Mass	E_U	$W_{1,h}$
RAI	17/36	17/36	5.166	6.774	0.2870	0.1523
RAI-S	25/36	23/36	1.728	7.257	0.2330	0.1094
DOME	26/36	22/36	3.857	7.019	0.2591	0.1393
DOME-S	25/36	23/36	1.728	7.257	0.2330	0.1094

For RAI, the mean leakage change is -3.438 percentage points, with a descriptive 95% paired bootstrap interval $[-5.282, -1.926]$. For DOME, the change is -2.129 points, with interval $[-3.155, -1.268]$. The corresponding clean prediction changes are -0.0540 $[-0.0800, -0.0325]$ and -0.0261 $[-0.0338, -0.0181]$. These intervals summarize the saved 36-case panel under the stated resampling convention; they do not adjust for multiple reported endpoints or establish external experimental performance.

RAI-S and DOME-S yield effectively identical primary summaries to the displayed precision. Their common post-proposal objective, support rule, and refinement can erase differences between proposal paths. These outcomes support the shared selection procedure and give no evidence that two independent improvements have been discovered.

7.2. Scenario-Level Failures and a DOSY Illustration

Figure 2 separates ordinary mixtures from difficult identification regimes. Both screened methods recover order and all rates in all six one-, two-, and three-separated-component cases. In close-rate cases, they recover the correct order in 5/6 problems, but all rates in only 4/6. For proportional spectra, they recover neither the full three-rate structure nor the correct order in any of six cases. The original DOME recovers the correct order in one of these cases, so the screened version has a visible regression.

In the weak family, the correct order is selected in 2/6 cases, but the weakest rate is matched in only 1/6; neither original method matches it in these six cases. In the one matched screened case, approximately 72% of the true minority spectral mass lies within the retained allowed support. The support metric is a recall of permitted spectral channels, not a claim of unbiased amplitude recovery. These results leave weak-species detection unresolved.

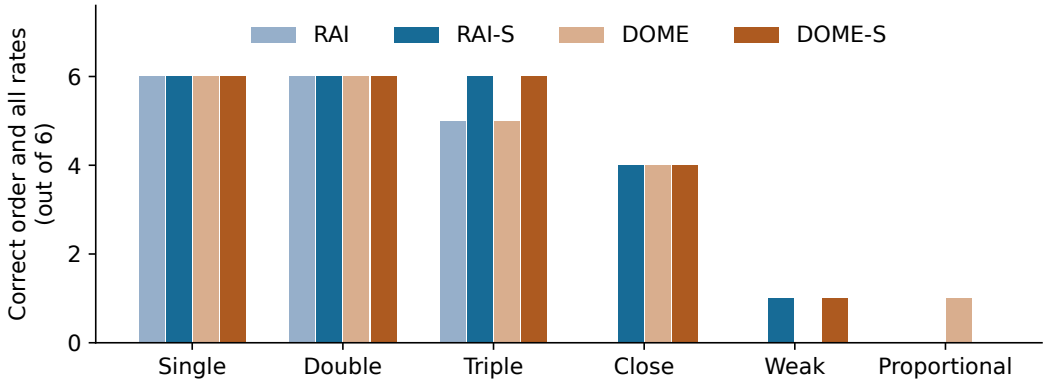


Figure 2. Recovery of both order and all rates by scenario in the new atomic test. Every bar has six independently generated problems. Matching uses a log-rate tolerance of 0.25. Narrow output bands and correct order alone do not imply successful localization.

Figure 3 provides one fixed illustration, rather than selecting a new representative after ranking the test outputs. The ground truth is displayed explicitly. Correlation is used

through common rates and component-specific spectral amplitudes; it is not an assumption that every spectral channel has the same decay. The color threshold is a plotting convention applied to every panel and has no role in model order or support estimation.

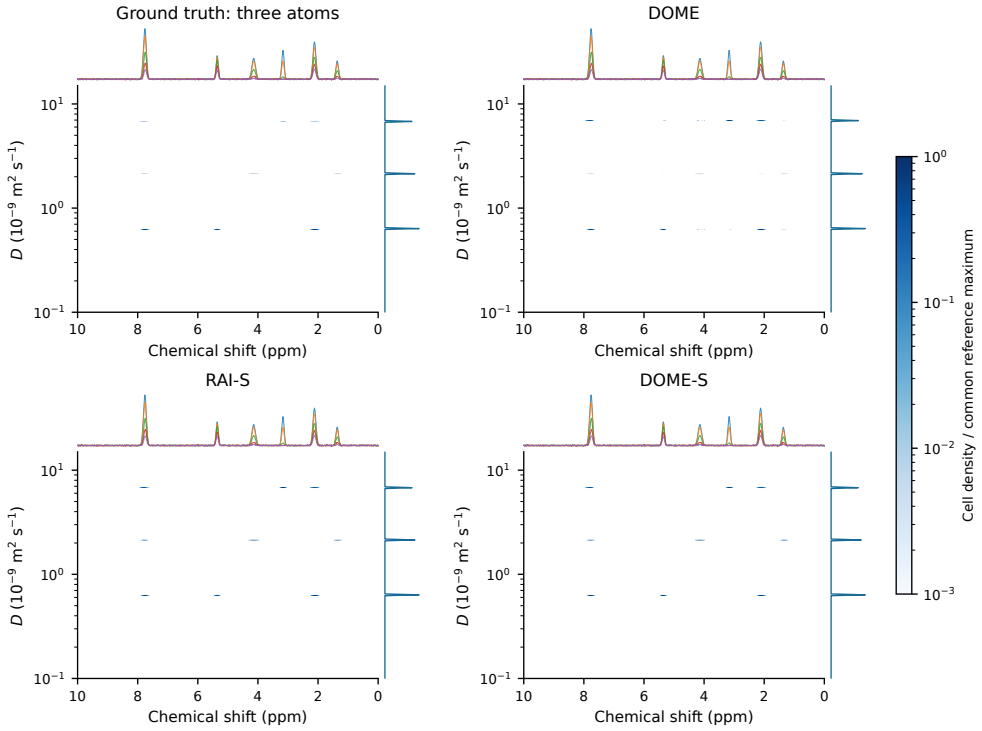


Figure 3. Saved independent test problem 18, with three true atomic rates and SNR 200. The complete reference is shown alongside DOME and the two support-selected outputs. All panels use the same selected spectral channels, 256-bin mass-to-density display, and common absolute reference maximum; values below 0.1% of that maximum are not colored. The top traces show measured attenuation spectra. Side projections have separate horizontal scales. The display creates finite-width cells for atoms; it is not a recovered continuous width. Quantitative results use the unthresholded saved outputs. Color denotes cell density divided by the common reference maximum on a logarithmic scale; darker blue indicates greater density. White areas lie outside the displayed support or below the stated display floor.

7.3. Null Controls, Broad Distributions, and Transfer

All four atomic methods select the null model in all six noise-only controls. This observed count is not a guarantee of a population false-positive rate: even under a common Bernoulli model, zero false selections in six trials gives a one-sided 95% upper bound of $1 - 0.05^{1/6} = 0.393$. The larger revision controls are reported separately in Appendix F, without pooling them with the frozen study. On the six broad controls, the screened methods have clean prediction error 0.2382, compared with 0.3664 for RAI and 0.2837 for DOME. Their binned shape score is 0.1638, compared with 0.2686 and 0.1972. An atom model nevertheless cannot reproduce nonzero distributional width exactly. The apparent improvement in prediction is therefore not evidence of recovered width. For a positive diffusion measure with mass M, mean $\bar{D}$, variance v and support $D \geq D_{\min} \geq 0$, convexity and Taylor's theorem give the finite-measurement bound

$$0 \leq \int e^{-bD}\, d\mu(D) - Me^{-b\bar{D}} \leq \tfrac{1}{2} Mb^2 e^{-bD_{\min}} v.$$

Thus, a narrow continuous profile can have an attenuation close to an atomic one on the sampled interval. Reducing the allowed spectral support can trade approximation

bias against estimation variance. This explains why lower prediction error is compatible with an incorrect measure class; it is not a proof that variance reduction alone caused the observed improvement or that the true distributional width was identified. Counting its fitted rates against the number of generating broad profiles is a descriptive center-count comparison, not a valid literal atom-recovery claim.

On the 24 archived finite-width problems, leakage decreases from 1.245% to 0.082% for both native paths. Clean prediction error changes from 0.2650 to 0.2329 for RAI and from 0.2492 to 0.2329 for DOME. The selected-order count agrees with the number of generating profiles in 19/24 screened fits versus 18/24 native fits, while the stricter matching count remains 18/24. Transfer performance is encouraging for spectral assignment, but the differing true measure class prevents pooling it with the atomic test.

Figure 4 retains the existing TRAIn-MF reconstruction and its native grid. For this case, the native atomic leakage of approximately 1.975% decreases to 0.279% after support selection, with three selected rates. The TRAIn-MF result is used to display the distinction between distributional factorization and atomic inference. It was not recomputed for this manuscript, and the figure is not a new all-method performance ranking. No conclusion about polymer molecular weights is drawn without a separate calibration and experimental analysis.

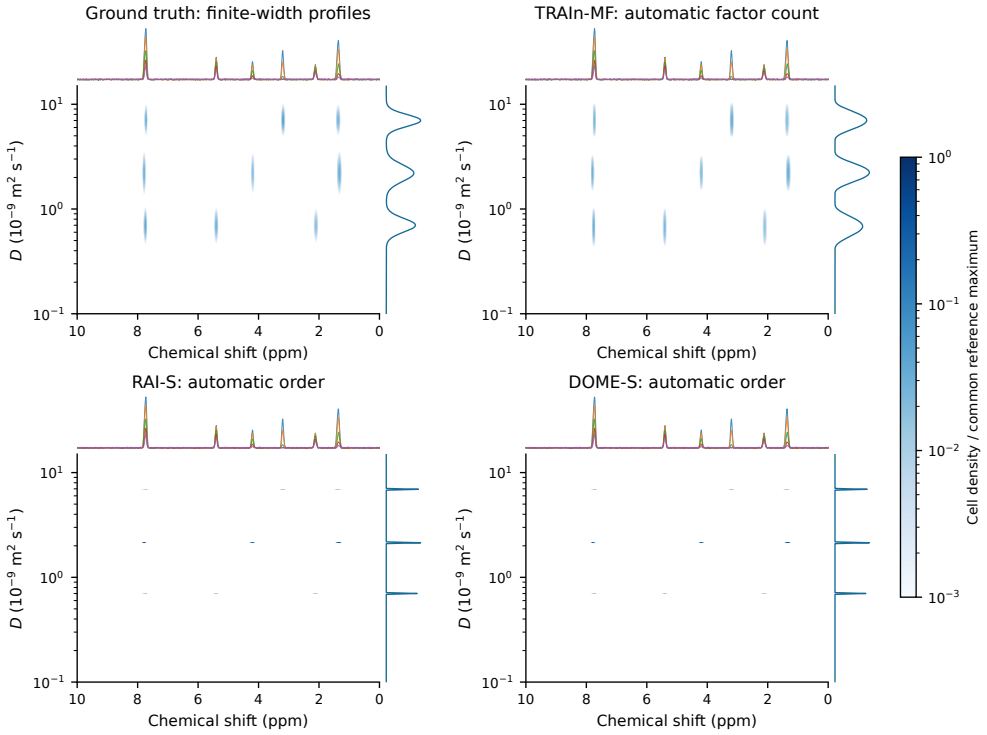


Figure 4. Archived three-component problem at SNR 200 (replicate 1): three finite-width lognormal diffusion profiles, SNR 200, and 120 selected spectral channels of 1024. Ground truth, the unchanged automatic TRAIn-MF reference, RAI-S, and DOME-S are shown. TRAIn-MF uses its original masses and original cell edges without remapping. Its factor count and the atomic methods' order are selected automatically but have different meanings. The common reference maximum is 63.34986357 in the saved cell-density units, with a 0.1% display floor. Side projections are separately scaled. Fitted atoms occupy individual export cells and do not estimate the true widths. Color denotes cell density divided by the common reference maximum on a logarithmic scale; darker blue indicates greater density. White areas lie outside the displayed support or below the stated display floor. The colored top traces correspond to individual acquisitions.

7.4. Exploratory Neural Policy Comparison

Table 5 reports the policy comparison. On 24 new test problems, the relative change in mean binned shape score is approximately 0.041%, with mean paired difference -0.000111 and bootstrap interval $[-0.001356, 0.001105]$. The clean-error interval also includes zero. On the archived transfer panel, the shape score improves by approximately 1.48%, but this does not establish an advantage on the new test. One fit per policy in the archived panel fails its numerical convergence criterion; both outputs remain in the reported averages.

Table 5. RAI-Net policy comparison on its predefined panels. These are distributional methods, so no chemical order recovery count is assigned. All saved outcomes, including the two numerical failures, enter the averages.

Panel	Policy	$W_{1,h}$	E_U	Numerical Success
New test, $n = 24$	Original	0.272268	0.307991	24/24
	Retrained	0.272157	0.307921	24/24
Archived, $n = 24$	Original	0.204769	0.326401	23/24
	Retrained	0.201739	0.326324	23/24

The new neural panel includes atomic, broad, and null cases, with zero shape-score contributions for nulls under the archived bookkeeping convention. Its averages therefore do not share the population of Table 4. The policy ranks adaptive actions within an existing numerical procedure. It neither adds observations nor certifies the chemical support. These data do not support promoting the retrained policy as a superior substitute for the support-selected atomic estimator, and the two are not compared as if they shared the same output class. This small exploratory comparison is inconclusive. A separate experiment uses five reproducible training seeds and 120 new whole-problem tests, including noise and acquisition-design shifts, against a classical policy with the same search caps (Appendix G.4). Numerical failures, realized costs and paired whole-problem intervals are reported there; the extended experiment is not pooled with Table 5.

7.5. Extended Computational and Experimental Validation

The factorial experiment in Appendix G examines screening and predictive selection separately. For RAI, screening at native order reduces clean error from 0.30677 to 0.29087 noise units. Predictive selection alone gives 0.26189, and combining the stages gives 0.23145. DOME reaches the same final factorial result from a different native-order baseline. These interventions support a contribution from both shared stages on this panel, with an interaction, without demonstrating an independent advantage of either proposal family. Unlike the mandatory-screening factorial arm, the production selector also considers unrestricted support and reproduces the original saved result.

In the larger weak-component panel, RAI-S matches the minority rate in 45/96 fits and DOME-S in 44/96. Figure A2 shows how recovery changes with SNR and amplitude. Even though all fits meet their numerical criteria, the smallest components are often missed; a nonselected component cannot generally be declared absent. The TRAIn-MF study in Appendix G.1 also shows why iteration counts need a separate stationarity check. Only 14 of 36 selected fits meet the 10^{-6} KKT tolerance within the common SQP budget. We retain the 22 unresolved fits in the objective and prediction summaries.

All 720 final outputs in the larger neural study pass their numerical checks. After averaging the five neural metrics within each problem, the difference from the classical shortlist is $\Delta W_{1,h} = -0.00054$, with a descriptive 95% interval $[-0.00174, 0.00056]$ over 105 signal-present problems. The clean-error difference is -0.00012, with interval $[-0.00044, 0.00010]$ over all 120 problems. Both intervals include zero, so the study does not establish improved

accuracy. The neural policies require fewer fixed-tree fits under the same search caps; Appendix G.4 gives the cost definitions and accounts for the reused training traces.

Table A8 compares the attributed implementations of SCORE and LOCODOSY, a positive-amplitude SCORE variant, and positive distributional controls. All methods predict the same withheld observations, but their component counts and distributional shape scores have different meanings. Every returned estimate enters the prediction summary; numerical failures and outputs that do not define a positive measure are identified separately. This is a comparison of the stated implementations and tuning policies, rather than a universal ranking of algorithm names.

The industrial resin study in Appendix G.6 tests the methods on measured spectra with processing distortions and strongly correlated background. Here, the fitted atomic order describes a model of the data; reaching the search cap does not identify the number of constituents. A signed fit can have a smaller predictive error while relying on nearly coincident rates and canceling spectra. These experimental outcomes test the scope of the synthetic conclusions and retain model mismatch and optimization flags.

The public 30-acquisition example in Appendix G.7 provides a useful counterexample. Its archived sample text documents propanol and sucrose. All four final global fits and all 27 local SCORE solver calls pass their numerical checks, yet signed SCORE predicts best: test NRMSE is 0.232%, compared with 0.582% for RAI-S and DOME-S and 0.270% for LOCODOSY. The signed SCORE fit also passes the conditioning checks, so its advantage cannot be attributed to failed optimization. Amplitude constraints and possible acquisition-model error affect this comparison. RAI-S and DOME-S reach the cap and flag model mismatch. We therefore find no general experimental predictive advantage for the positive global selector. The nominal solutes do not establish the full composition, chemical assignments or independently measured diffusion coefficients.

7.6. Numerical Diagnostics

Across 144 support-selected fits in the primary and transfer panels, the maximum relative amplitude KKT residual was 1.12×10^{-15} and the maximum normalized projected log-rate residual was 1.34×10^{-7}. Four poor-precision flags and four atomic model-mismatch flags were recorded. In the independent 100-column NNLS comparison, the relative prediction discrepancy was 4.62×10^{-17}. All native and support-selected atomic fits in these panels met their numerical stopping criteria. The two neural failures occurred on the same problem. Appendix G gives separate numerical failure counts for the additional studies. A small stationarity residual does not establish precise rates or chemical identification.

8. Discussion

Our formulation makes clear which part of the result comes from the representation, the regularization, the optimizer and the available information. Exporting a fit on 256 bins reduces display quantization; it does not create 256 independent diffusion observations. Likewise, a lower prediction error cannot by itself identify distributional width, chemical species or a unique factorization.

The support selector limits the assignment of a proposed rate to unrelated spectral regions. It first asks whether that rate contributes beyond the other candidate decays, using groups of channels to retain coherent weak signals. Unrestricted support remains a candidate. The factorial study finds contributions from both screening and predictive selection, together with an interaction. It does not establish an independent advantage for the RAI or DOME proposals. Their primary outputs coincide, as do many of the additional outputs, although the weak-component panel contains one different minority-rate match. The searches have not been proved equivalent.

The spectral groups are geometric windows and need not coincide with chemical resonances. Masking can remove weak signal, errors in nuisance rates can bias the residualized score, and nearly coincident rates make its direction ill-conditioned. Proposition 7 and the stability bounds describe fixed designs. The 300 null matrices and 96 weak-component problems characterize particular generators; they do not calibrate $\alpha = 0.01$ as a universal family-wise error rate. The prediction allowance uses the correct difference between fitted predictions, but choosing the best model and searching supports make its subsequent use heuristic. The discovery masks in the new weak-component and experimental studies exclude validation observations. This removes one source of data reuse, while leaving uncertainty in the estimated nuisance model and the need for selection-adjusted inference.

A TRAIn-MF factor represents a shared distribution, which may be broad or multimodal. Its unit-mass normalization fixes scale, not chemical identity. The runtime curvature check concerns injectivity of a conditional subproblem. The controlled warm-start study supplies factor count to isolate optimization and records nonstationary outcomes. It therefore tests neither automatic species selection nor convergence to a global minimum. The search caps and iteration budgets remain essential parts of the reported procedure.

An atomic model can predict the measured attenuation of a broad distribution well because the Laplace operator smooths differences between measures. The bound in Section 7.3 explains why good prediction can coexist with an incorrect distributional model. Screening may reduce estimation variance at the cost of approximation bias, but the prediction gains alone do not isolate that mechanism or recover the true width. This matters for industrial resins, whose heterogeneity may require more than a few atoms. Their selected model order should not be read as a count of molecular constituents.

The external comparisons show how strongly the result depends on the fitted representation and the chosen evaluation measure. Signed cancellation, native order rules and incomplete optimization remain visible alongside prediction error. A signed fit may not define a positive measure, so transport scores computed only on eligible outputs describe different subsets. On the public mixture, signed SCORE converges and passes the stability checks; it and LOCODOSY predict better than the positive global estimators. We report that advantage without treating it as proof of chemical recovery. The nominal composition provides neither a complete reference order nor independently measured diffusion coefficients. These results also cannot rank the four external methods that were not run.

The larger neural study compares five seeded policies on 120 new problems, including changes in noise and acquisition design. All policies use the same search caps and numerical acceptance checks. The policies share a trace bank and test problems; their 720 fitted outputs are not 720 independent mixtures. We average seed-level metrics within each problem before the paired analysis and retain numerical failures. The pooled accuracy intervals include zero. The cost reduction is conditional on the stated training and search design and does not establish an end-to-end speedup or experimental chemical recovery.

The resin measurements provide four specimen series, each with three diffusion-time blocks. They do not provide twelve independently prepared mixtures of known composition. Their strongly correlated background and possible phase, baseline, suppression, exchange, relaxation or encoding errors can violate the ideal observation model. Support is detected without the reserved acquisitions; negative coherent structure and unestimated regions remain documented. Prediction on reserved acquisitions and consistency across diffusion times are useful checks, but neither supplies a component diffusion reference or a molecular-weight calibration. The experimental interpretation must remain within those limits.

9. Conclusions

Joint positive Laplace inversion can improve spectral assignment when shared components are supported by the data. The distinction between bin masses, densities, atoms and shared profiles is essential to interpreting the result. Our conditional stability and uniqueness calculations explain where reconstruction can fail; the additional experiments show those limits in practice. Weak rates are missed, finite optimization budgets leave some fits nonstationary, and the neural accuracy gains remain inconclusive. The observed reduction in spectral leakage therefore supports a specific empirical contribution. Chemical identity and distributional width require additional evidence.

Supplementary Materials: The following supporting information is available at https://www.mdpi.com/article/10.3390/math1010000/s1: Supplementary File S1, *Computational protocols and data*. It contains the study designs, numerical settings, model outputs, trained neural policies and processed experimental inputs used in the comparisons.

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Data Availability Statement: The method implementations, synthetic benchmark generators and public validation summaries are available in the TRAIn-DOSY repository at https://github.com/fmarrabal/train-dosy (accessed on 9 November 2026) [50]. The processed industrial resin spectra and fitted amplitudes are supplied in Supplementary File S1 and are not distributed in the public repository. The public DOSY mixture is available from the Manchester NMR Methodology Group [51]; its preprocessing and acquisition partitions are documented in File S1.

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Appendix A. Further Mathematical and Numerical Details

Appendix A.1. Scaling and Invariance

In the atomic model, replacing (Y, σ) by $(cY, c\sigma)$ for $c > 0$ maps A to cA, leaves rates fixed, and preserves the normalized objective and screening scores. The same transformation preserves the paired allowance in Equation (44). Thus, for unchanged candidate paths and masks, the exact-arithmetic decision is invariant to signal units. Finite solver tolerances and tie handling can introduce small numerical differences. Changing physical diffusion units with the reciprocal change in b also preserves bD and the predictions. Rescaling observations without rescaling σ changes the statistical problem.

The TRAIn-MF objective has a related property through $\widetilde{Y} = Y/s_Y$: multiplying the data by c changes s_Y by c, leaves normalized factors unchanged, and rescales the exported amplitudes. Changing the number of spectral channels is different. Averaging the data term by p and the amplitude penalty by p does not make arbitrary deletion of spectral channels innocuous; the available shared information changes. The exact grid, quadrature, and regularization operator must also be kept fixed when claiming numerical equality of two TRAIn-MF outputs.

Appendix A.2. A Local Separation Expansion

For the split in Equation (24), write $\delta_1 = -(1-\pi)\Delta$, $\delta_2 = \pi\Delta$. Taylor expansion of $e^{-t(m+\delta)}$ in the dimensionless coordinates of Section 4.7 gives a first-order term proportional to $\pi\delta_1 + (1-\pi)\delta_2 = 0$, and a second-order term proportional to $\pi\delta_1^2 + (1-\pi)\delta_2^2 = \pi(1-\pi)\Delta^2$. The bounded-interval remainder is $O(\Delta^3)$ for a fixed finite acquisition design. The derivative in Δ vanishes at collision; a variance parameter can expose a nonzero second-order direction. This changes search geometry, not the measured information. After NNLS and rate refinement, moment preservation of the proposal is not a constraint on the final estimate.

Appendix A.3. Support Geometry and Precision Warnings

For an interior fixed active set, the first-order rate information after treating amplitudes as nuisance involves projected kernel derivatives, weighted by amplitudes across spectral channels. Extra rows or differently weighted spectra can add information. In proportional spectra, repeated columns may reduce variance by replication, while preserving the same decay-mixture proportions. If all relevant projected derivatives vanish, local rate separation fails irrespective of the plotted bin count.

The software additionally records the condition of the full residual Jacobian and the diagonal of a pseudoinverse of its Gram matrix. Residual-dependent terms, singular directions, active-set changes, and model selection prevent interpreting this output as a calibrated Fisher-information confidence interval. In particular, a pseudoinverse can return finite entries even when some directions are unidentifiable. The separate condition warning must be retained. A small normalized projected gradient establishes neither small parameter error nor precise component identification.

Appendix B. Effect of Including Null Controls in Shape Metrics

Table A1 averages over 42 atomic problems, including six null cases assigned zero leakage and shape error by convention. Including these zeros lowers the average without changing any reconstruction. The primary analysis therefore reports shape scores for the 36 signal-present problems and reports null selection separately.

Table A1. Earlier 42-case convention, including six null controls. Columns are defined as in Table 4. The zero-valued null shape contributions are bookkeeping conventions, not normalized Wasserstein distances.

Method	Correct Order	Order and Rates	Leakage	Unmatched Mass	E_U	$W_{1,h}$
RAI	23/42	23/42	4.428	5.806	0.2460	0.1306
RAI-S	31/42	29/42	1.481	6.220	0.1997	0.0938
DOME	32/42	28/42	3.306	6.016	0.2221	0.1194
DOME-S	31/42	29/42	1.481	6.220	0.1997	0.0938

Appendix C. Computational Reproducibility

The joint atomic reconstructions were implemented in Python. TRAIn-MF was run in MATLAB R2026a Update 2 with Optimization Toolbox 26.1. The public repository [50] provides the synthetic data generators, acquisition partitions, model settings and evaluation code. Supplementary File S1 gives the study-specific protocols and processed experimental inputs. The primary DOSY figures use the fitted profiles or atomic amplitudes from the reported experiments; drawing these maps does not require estimating the models again.

The five RAI-Net policies were trained on CPU. File S1 specifies the training and test problems, the fitted neural policies and the corresponding classical control. Unresolved numerical fits are included in the reported results.

Appendix D. Additional Inversion Models

Appendix D.1. The Density-RAI Graph

For column ℓ, let $a_\ell = \max_i |y_{i\ell}|$, $v_{i\ell} = y_{i\ell}/a_\ell$, and $s_{i\ell} = \sigma_{i\ell}/a_\ell$, with a positive numerical floor on a_ℓ. For two columns meeting the SNR requirement, the inspected density implementation uses

$$d_{\ell m} = \frac{1}{n} \sum_i \frac{(v_{i\ell} - v_{im})^2}{s_{i\ell}^2 + s_{im}^2 + v_{i\ell}^2 \max_k s_{k\ell}^2 + v_{im}^2 \max_k s_{km}^2}. \tag{A1}$$

An edge is admitted only below the fixed distance threshold and when both columns include the other among their nearest eligible neighbors; ties at the neighbor cutoff are retained. Then $W_{\ell m} = e^{-d_{\ell m}}$. The denominator accounts heuristically for uncertainty in shape normalization, but Equation (A1) is not a calibrated test statistic. Estimating this graph from the available fitting rows changes the estimator; fixed-graph convexity applies conditionally after that construction.

Appendix D.2. A Feasible Primal–Dual Gap for Adaptive Haar RAI

Use the notation of Equation (19), and let $\mathcal{P}(Z) = \lambda\sqrt{p}\sum_{h\geq 2} \|Z_{h,:}\|_2$. Choose $N \leq 0$ entrywise and a matrix D with first row zero and $\|D_{h,:}\|_2 \leq \lambda\sqrt{p}$ for detail rows. These are dual-feasible for the nonnegative cone and Haar penalty, respectively. Let $u_\ell^* = H_\ell^{-1}(c_\ell - N_\ell - Q^\top D_\ell)$. Then

$$\mathcal{G}(U, N, D) = \frac{1}{2}\sum_\ell \|U_\ell - u_\ell^*\|_{H_\ell}^2 + \mathcal{P}(QU) - \langle D, QU\rangle - \langle N, U\rangle. \tag{A2}$$

For $U \geq 0$, every term is nonnegative, and completing the quadratic square shows that $\mathcal{G}$ equals a primal value minus the corresponding feasible dual lower bound. It therefore bounds suboptimality on this fixed tree. In ADMM the code uses the feasible primal iterate V, projects ρd_V to the nonpositive orthant, and projects ρd_Z to the row balls above. For the separate penalty, the balls become entrywise intervals $[-\lambda, \lambda]$. At $\lambda = 0$, it uses $N = \min(-\nabla f(U), 0)$ and $D = 0$ after ridge NNLS. Floating-point gap values are floored at zero to remove negative roundoff; this is a numerical bound construction, not interval arithmetic.

For completeness, let $m_{j\ell}$ be the current mass in leaf j, $\bar{k}_j$ its unit-mass kernel, and w_j its log width. At each spectral channel, the nuisance matrix concatenates columns $m_{j\ell}\bar{k}_j$ and $(m_{j\ell}w_j/2)\partial_x k(x_j)$ for occupied leaves, divided by that spectral channel's noise. Occupied

means $m_{j\ell} > \max(0.02 \max_h m_{h\ell}, \text{median}_i\, \sigma_{i\ell})$. Its left singular vectors with singular value at least one form U_{vis}. For a prospective split contrast $t_j = (\bar{k}_{j,L} - \bar{k}_{j,R})/2$, put

$$v_{j\ell} = (I - U_{\text{vis}} U_{\text{vis}}^{\top})(t_j/\Sigma_\ell), \quad s_{j\ell} = \frac{\langle v_{j\ell}, (Y - B_{\text{fine}} C)_\ell/\Sigma_\ell \rangle^2}{\|v_{j\ell}\|_2^2}, \quad I_{j\ell} = m_{j\ell} \|v_{j\ell}\|_2. \tag{A3}$$

The score is set to zero if the denominator is at most 10^{-24} or $I_{j\ell} < 0.05$. A raw score $s^0_{j\ell}$ omits nuisance projection. Joint-detail proposals rank leaves by $\sum_\ell s_{j\ell}$, private-detail proposals by $\max_\ell s_{j\ell}$, position proposals by $\sum_\ell s^0_{j\ell}$, and occupied-region proposals by $w_j \max_\ell m_{j\ell}$. At most four leaves are split per action, subject to capacity. The sibling pair with the smallest difference in its regional mass vectors supplies a merge proposal. These are deterministic heuristics inside the finite tree search, not tests of chemical identity.

Appendix D.3. The 14 RAI-Net Action Features

For an action affecting leaves I, current leaf count m, candidate count m', and fine count q, the feature vector used in Equation (21), in stored order, is

$$\begin{aligned} \phi_a = \big(& m/q,\ m'/q,\ \mathbf{1}_{\text{merge}},\ \mathbf{1}_{\text{position or occupied}}, \\ & \log(1 + \textstyle\sum_{j\in I,\ell} s_{j\ell}),\ \log(1 + \max_{j\in I,\ell} s_{j\ell}),\ \log(1 + \sum_{j\in I,\ell} s^0_{j\ell}), \\ & \log(1 + \max_{j\in I,\ell} I_{j\ell}),\ \frac{\sum_{j\in I,\ell} m_{j\ell}}{\max(\sum_{j,\ell} m_{j\ell}, 10^{-300})},\ \frac{|I|^{-1} \sum_{j\in I} w_j}{\sum_j w_j}, \\ & \log(1+\lambda),\ \log(1 + n_{\text{supplied}}),\ \log(1+p),\ \log(1 + F_{\text{current}}) \big). \end{aligned} \tag{A4}$$

The acquisition-count feature is the number of supplied reconstruction rows before the internal split, not an external-test count. The action scorer sees the state derived from fitting rows; validation is used separately to train the revised priority target and to select completed penalty paths. Networks and checkpoints are therefore tied to the feature semantics as well as their matrix dimensions.

Appendix D.4. DOME Support Averaging

The historical averaging variant holds the rates selected by native DOME fixed. At each spectral channel it enumerates subsets J of these rates, including the empty subset, and computes NNLS amplitudes a^J_ℓ and $c_{J\ell} = \text{RSS}_{J\ell}/\sigma^2 + 6|J|$. The penalty counts the nominal subset size even if some fitted amplitudes are zero. With temperature $T = 0.5$, it solves the finite entropy-regularized averaging problem

$$\min_{w \ge 0, \sum_J w_J = 1} \sum_J w_J c_{J\ell} + T \sum_J w_J \log w_J, \quad w_{J\ell} = \frac{e^{-c_{J\ell}/T}}{\sum_L e^{-c_{L\ell}/T}}, \quad \bar{a}_\ell = \sum_J w_{J\ell} a^J_\ell. \tag{A5}$$

Soft weights can retain weak contributions; they are not posterior probabilities or confidence levels. The rate set is not re-estimated, and single-channel inputs bypass this averaging. This construction is mathematically different from DOME-S, which screens allowed supports, refines rates for each candidate, and selects by reserved-acquisition prediction.

Appendix D.5. RAI-FLEX: Joint Atomic and Smooth Profiles

RAI-FLEX uses r_a atoms and r_c shared smooth profiles. Let φ_h be 24 nonnegative, unit-integral cubic B-splines in log diffusion, G their integrated Laplace matrix, and L_2

their second-derivative quadrature operator. The predicted matrix is $H(d, C)A$ with $H(d, C) = [K(d), GC]$. The objective is

$$\min_{d, C \geq 0, \mathbf{1}^\top C = \mathbf{1}^\top, A \geq 0} \tfrac{1}{2}\|(H(d, C)A - Y)/\Sigma\|_F^2 + \tfrac{\lambda}{2}\|L_2 C\|_F^2, \quad \lambda = 0.01. \tag{A6}$$

Rates satisfy the physical bounds. NNLS eliminates A at every evaluation. Initial optimization uses log rates and columnwise $C = \mathrm{softmax}(Z)$ with bounded logits, followed by direct simplex refinement to reach boundary profiles that a softmax gradient can represent poorly. The profile gradient is

$$G_C = G^\top((HA - Y)/\Sigma^2)A_c^\top + \lambda L_2^\top L_2 C, \qquad G_{Z,j} = C_j \odot \{G_{C,j} - (C_j^\top G_{C,j})\mathbf{1}\}. \tag{A7}$$

Here A_c denotes the continuous-profile rows. A small logit gradient alone need not imply a small simplex KKT residual. After L-BFGS-B, up to 40 refinement sweeps alternate convex simplex quadratic profile updates, bounded atomic log-rate updates, and NNLS amplitudes, recording a scaled KKT residual with threshold 10^{-6}.

The archived comparison searches allocations $r_a + r_c = r$ for $r = 1, \ldots, 4$; the general software default has a larger cap. There is no zero-model candidate in this RAI-FLEX search. A local effective dimension projects profile/rate tangent directions away from active spectral-amplitude directions. If $T_\perp$ is this weighted tangent matrix and P the tangent penalty Hessian, the count is

$$d_{\mathrm{eff}} = d_A + \mathrm{tr}\{(T_\perp^\top T_\perp + P)^+ T_\perp^\top T_\perp\}. \tag{A8}$$

Here, $d_A = \sum_\ell \mathrm{rank}(\mathrm{diag}(\Sigma_\ell^{-1})H_{J_\ell})$ for the active amplitude subsets J_ℓ; density tangent contrasts use spline coefficients above 10^{-5}. The pseudoinverse is numerically truncated. The selector minimizes $\mathrm{RSS}/\sigma^2 + 2d_{\mathrm{eff}} + 2d_{\mathrm{eff}}(d_{\mathrm{eff}} + 1)/\max(1, np - d_{\mathrm{eff}} - 1)$ over finite attained scores. It records stationarity but does not exclude every nonstationary candidate from this native selection. The count ignores active-set selection and nonlinear curvature and is therefore heuristic. Competing allocations within two score units are marked as type-ambiguous. The method does not certify that a fitted atom is a discrete molecule or that a broad profile is a polymer; these remain representation choices.

Appendix D.6. Partial-C: One Shared Profile and Unrestricted Private Densities

The generalized partial-C comparator instantiates Equation (5) with 24 unit-integral cubic B-splines, one shared profile s, $C = sa + V$, $a, V \geq 0$, and $\mathbf{1}^\top s = 1$. Its penalty operators approximate the first derivative of the total density and the L^2 norm of the private density: $L^\top L = Q$ and $F^\top F = G$ in Section 3.4. These are different from RAI-FLEX's second-derivative penalty on its shared profiles. It fixes $\lambda_s = 0.01$ and $\lambda_p = 10$.

For each proposed s, Equation (6) gives the joint conditional NNLS fit of (a_ℓ, v_ℓ), including the private penalty. With these amplitudes profiled out, the envelope gradient is

$$\nabla_s J = \{K^\top(KC - Y)/\sigma^2 + \lambda_s L^\top L C\}a^\top/p. \tag{A9}$$

Two observation-derived starts are refined by simplex-constrained SLSQP, with at most 300 iterations, and the lowest attained objective is retained. Shared-profile and conditional-linear KKT residuals at most 10^{-5} define its stationarity diagnostic. Proposition 5 proves conditional uniqueness under its stated conditions; it does not prove uniqueness of s or of a physical decomposition. Shared rank is fixed to one in this comparator, not automatically selected. Because V can represent every positive basis density, a rank-one shared term places no rank-one restriction on the complete signal.

Appendix D.7. CIRCE Reconstruction Adapters: Positive Quadratics and a Feasible-Set Check

The CIRCE models considered here use a common basis for the benchmark acquisition design. Its 256 normalized piecewise-linear hats give masses $C \geq 0$, operator K, and density derivative R. Let a_h be the unnormalized hat areas and $J_a = \operatorname{diag}(1/a_h)$. Normalize observed columns to $f_\ell = Y_\ell / \max(\|Y_\ell\|_2, \sqrt{n}\sigma)$ and form

$$W_{\ell m} = \frac{\exp(-\|f_\ell - f_m\|_2^2/0.1)}{q \max(1, p-1)} \ (\ell \neq m), \quad W_{\ell\ell} = 0, \qquad L_g = \operatorname{diag}(W\mathbf{1}) - W. \tag{A10}$$

The full-basis model minimizes the convex quadratic

$$F_1(C) = \tfrac{1}{2}\|(KC - Y)/\sigma\|_F^2 + \tfrac{0.02}{2}\|RC\|_F^2 + \tfrac{0.01}{2}\operatorname{tr}(J_a^2 C L_g C^\top), \qquad C \geq 0. \tag{A11}$$

The positive-decoder model uses $C = TZ$, $Z \geq 0$, where T is a fixed positive mass-preserving decoder:

$$T_{hk} = \frac{a_h \exp\{-(x_h - x_k)^2/(2w^2)\}}{\sum_j a_j \exp\{-(x_j - x_k)^2/(2w^2)\}}, \qquad w = 2\,\mathrm{mean}_h(x_{h+1} - x_h). \tag{A12}$$

With $s_\ell = \max(\max_i |Y_{i\ell}|, 5\sigma)$, its objective is

$$F_2(Z) = \tfrac{1}{2}\|(KTZ - Y)/\sigma\|_F^2 + \tfrac{0.2}{2}\sum_\ell \frac{\|RTZ_\ell\|_2^2}{s_\ell^2} + \tfrac{0.01}{2}\operatorname{tr}(J_a^2 TZ L_g Z^\top T^\top). \tag{A13}$$

The decoder is part of the estimator, not display smoothing; its positive image is a restricted cone within the original basis representation. There is no learned decoder in this comparator. Both objectives have gradients given by the corresponding quadratic matrices, with $T^\top$ applied in the positive-decoder model. The usual implementation uses projected FISTA with restart and a fixed Hessian-norm Lipschitz bound. A "strict" configuration uses cyclic exact NNLS column blocks instead, including each graph column's off-diagonal contribution in the block right-hand side. Tolerances 10^{-6} and 10^{-10} distinguish numerical solves of the same objective, not additional physical inversion models.

The adapter then checks the returned physical masses against a residual cone and mass caps. Let e_h bound each basis column's quadrature error uniformly across acquisitions, using 64 composite-trapezoid subdivisions per segment, and $\varepsilon = \sqrt{\chi^2_{np,0.95}}$. Its check is

$$\|(KC - Y)/\sigma\|_F + \sqrt{n}\|e^\top C\|_2/\sigma \leq \varepsilon, \qquad \mathbf{1}^\top C_\ell \leq \frac{[Y_{i_{\min},\ell} + \varepsilon\sigma]_+}{e^{-b_{\min} D_{\mathrm{cap}}}}. \tag{A14}$$

Here, $i_{\min}$ is the smallest acquisition coordinate and D_{cap} is the comparator's supplied last diffusion-grid rate. Quadrature and mass checks include recorded floating-point tolerances; no interval-arithmetic certificate is asserted. The positive quadratic is solved first and checked afterwards. If its minimizer is feasible for the additional restrictions, it also minimizes the quadratic over that restricted set; otherwise the adapter reports an unverified estimate. It does not solve a constrained rescue problem. Its grid-center mass cap and data-dependent graph must not be promoted to universal continuous-measure coverage guarantees.

These formulations concern CIRCE reconstruction, not bounds over alternative measures. CIRCE-Net was not evaluated: the available trained model does not match the acquisition design, spectral support and basis used here. Neither CIRCE formulation enters the primary support-selection comparison.

Appendix E. Conditional Uniqueness and Correlated-Noise Identities

Appendix E.1. What Curvature Regularization Guarantees for a TRAIn-MF Profile Update

The positive amplitude ridge in Equation (31) guarantees a unique conditional amplitude minimizer. The analogous claim for S needs an additional condition: its curvature penalty is semidefinite, not positive definite.

Proposition A1 (Sufficient condition for a unique conditional profile update)**.** *Fix A and $\lambda_S > 0$ in Equation (29). If the only matrix $U \in \mathbb{R}^{q\times r}$ satisfying*

$$KUA = 0, \qquad LU = 0, \qquad \mathbf{1}^\top U = 0 \tag{A15}$$

is $U = 0$, the simplex-constrained S subproblem has a unique minimizer.

Proof. For a feasible affine direction U, the second directional derivative is

$$\frac{d^2}{dt^2} F(S + tU, A) = \|KUA\|_F^2/p + \lambda_S \|LU\|_F^2. \tag{A16}$$

The stated condition makes this positive for every nonzero direction with $\mathbf{1}^\top U = 0$, so the objective is strictly convex on the simplex affine hull. The product of simplices is nonempty and compact, giving existence; strict convexity gives uniqueness. The condition is sufficient, not a necessary characterization of uniqueness at a boundary optimum. □

A degeneracy example shows why the condition matters. With two equal amplitude rows $A_{1,:} = A_{2,:}$, choose $v \neq 0$ such that $Lv = 0$ and $\mathbf{1}^\top v = 0$, and set $U = [v, -v]$. Such a v exists for the interior second-difference operator: L has at most $q - 2$ independent rows, and adding the single mass constraint still leaves a nontrivial null space. Then $UA = 0$ and $LU = 0$. At any strictly positive feasible S, sufficiently small t preserves nonnegativity of $S + tU$, while both the prediction and the full objective remain exactly unchanged. Thus, positive curvature regularization and simplex normalization can coexist with indistinguishable profile pairs. This is a counterexample to strict convexity of every conditional update, not a claim that the saved TRAIn-MF example has this degeneracy. It also explains why replacing a local KKT check by a statement of chemical uniqueness would be incorrect.

For the specified interior curvature operator, the condition has a simple reduction. Let $w_h > 0$ be its trapezoidal log-grid weights and put

$$\bar{x}_w = \frac{\sum_h w_h x_h}{\sum_h w_h}, \qquad v_h = w_h(x_h - \bar{x}_w).$$

Since a zero discrete second derivative means that the density is affine in x, $\ker L = \operatorname{diag}(w)\operatorname{span}\{\mathbf{1}, x\}$. Imposing zero mass leaves the one-dimensional space spanned by v. Every U satisfying $LU = 0$ and $\mathbf{1}^\top U = 0$ therefore has the form $U = vc^\top$. Consequently,

$$KUA = (Kv)(c^\top A),$$

and the injectivity condition in Proposition A1 is equivalent to $Kv \neq 0$ and $\operatorname{rank}(A) = r$. In exact arithmetic, an acquisition with $b_i > 0$ guarantees $(Kv)_i < 0$: the weighted covariance of the strictly increasing log nodes and the strictly decreasing values $e^{-b_i D_h}$ is negative. This reduction is specific to the stated interior curvature operator, not an assertion about arbitrary regularizers.

For the fit illustrated in Figure 4, feasibility and scaled KKT residuals were checked during optimization. Conditional injectivity was evaluated only at the final factors, using the

ratio of the smallest to largest amplitude singular value and the ratio $\|Kv\|_2/(\|K\|_2\|v\|_2)$. A ratio below the stated tolerance leaves the condition numerically unresolved. Neither ratio certifies exact rank in floating-point arithmetic or uniqueness of the outer factorization.

The two ratios for this rank-three fit were 0.641979 and 0.415756, respectively, above the tolerance 10^{-10}. These values support conditional injectivity at the final factor pair; they say nothing about earlier iterates. In the separate convergence study, the same condition was checked during fitting, including cases with duplicated amplitude rows or a constant kernel (Appendix G.1). This criterion applies to the stated curvature operator with positive profile regularization.

Appendix E.2. Fixed-Design Scores Under a Specified Covariance

The iid identities in Propositions 7 and 8 have covariance-aware forms. These clarify their assumptions; the frozen solvers do not estimate or apply the covariance corrections below. Throughout this subsection vec stacks columns, and the covariance is specified independently of the evaluated observations.

Proposition A2 (Gaussian linear screening and paired quadratic loss)**.** *Let $Y = M + E$, with $e = \operatorname{vec} E \sim N(0, \Omega)$. For fixed correct rates $M = KA$, a fixed spectral channel group G, its indicator $g_G \in \mathbb{R}^p$, and the residualized direction v_j of Equation (41), put $w = g_G \otimes v_j$. If $w^\top \Omega w > 0$, then*

$$\frac{w^\top \operatorname{vec} Y}{\sqrt{w^\top \Omega w}} \sim N\left(\frac{\|v_j\|_2^2 \sum_{\ell \in G} A_{j\ell}}{\sqrt{w^\top \Omega w}}, 1\right). \tag{A17}$$

For fixed predictions P, Q independent of E, a fixed symmetric weight matrix $W \succeq 0$, and $\delta = \operatorname{vec}(P - Q)$, let $L_W(P) = \|\operatorname{vec}(P - Y)\|_W^2$. Then,

$$\mathbb{E}[L_W(P) - L_W(Q)] = \|\operatorname{vec}(P - M)\|_W^2 - \|\operatorname{vec}(Q - M)\|_W^2, \tag{A18}$$

$$\operatorname{Var}[L_W(P) - L_W(Q)] = 4\delta^\top W \Omega W \delta. \tag{A19}$$

Proof. The screening numerator is a fixed linear functional of a Gaussian vector. Its mean follows from $v_j^\top K_{-j} = 0$ and $v_j^\top k_j = \|v_j\|_2^2$, and its variance is $w^\top \Omega w$. Expanding the two quadratic losses cancels $e^\top W e$; the random remainder is $-2\delta^\top W e$. Its mean is zero and its variance gives Equation (A19). □

If $\Omega = \sigma^2 I$, Equation (A17) reduces to Equation (42); taking $W = \sigma^{-2} I$ also recovers Equation (46). For separable covariance $\Omega = \Gamma \otimes \Sigma_b$, the screening variance factors as $(g_G^\top \Gamma g_G)(v_j^\top \Sigma_b v_j)$. The usual $\sqrt{|G|}$ signal gain therefore requires the independent-channel variance scaling used in the benchmark. For positive-definite Ω and generalized least-squares weight $W = \Omega^{-1}$, the paired variance becomes $4\delta^\top \Omega^{-1} \delta$.

Estimated rates, groups, covariances, and the best prediction selected from the same data remain random. Substituting them into these formulas does not create a selection-adjusted test. In particular, holding out acquisition rows need not make errors independent when acquisition noise itself is correlated across those rows. A covariance model and a validation design respecting that dependence would be needed for a new calibrated procedure. The fixed-design identities do not establish correlated-noise performance. The supplementary sensitivity experiment in Appendix F is reported separately from the primary iid benchmark.

Appendix F. Synthetic Controls for Model Selection

The screened estimators were applied to new null, ablation and distortion panels without tuning their parameters against the outcomes. Each null matrix has 32 acquisitions and

1024 spectral channels. Eight acquisitions are reserved for external testing; the remaining 24 are divided into 18 candidate-fitting and six validation rows. For the fixed-window branch, the interval $4 < \text{ppm} < 4.65$ is specified independently of the observations.

Appendix F.1. Supplementary Null and Mechanism Checks

The supplementary null panel contains 300 independently generated problems, with 100 at each known noise standard deviation 0.02, 0.01, and 0.005. Each screened method selected a nonzero model in two of the same 300 fixed-window problems. Both selected rates were at the lower allowed boundary. Under a common-event-probability binomial model, the pooled 95% interval would be $[0.081, 2.387]\%$. This is a descriptive reference, without a claim of coverage for heterogeneous event probabilities across noise scales. Table A2 instead reports exact intervals within each noise stratum. These are paired method outcomes, not 600 independent replicates, and do not establish a universal 1% full-pipeline error guarantee.

Table A2. Additional fixed-window null controls. RAI-S and DOME-S have identical event counts on paired inputs. Intervals are two-sided Clopper–Pearson intervals.

Known σ	Nonzero Outputs	95% Interval (%)
0.020	1/100	[0.025, 5.446]
0.010	1/100	[0.025, 5.446]
0.005	0/100	[0, 3.622]

The separate discovery branch selects no spectral channels in 291 problems; only nine proceed to inversion, all returning the null model. Zero reported components in 300 discovery pipelines must therefore be distinguished from zero nonzero fits in nine inversions. Treating each pooled denominator as a common-probability binomial sample gives two-sided 95% upper reference endpoints of 1.222% and 33.627%, respectively; these are descriptive pooled comparisons, with the same heterogeneity limitation as above.

The matched three-arm comparison in Table A3 uses 24 previously frozen signal-present problems, one common proposal bank per method on 18 rows, six validation rows, the same refinement tolerances and a final 24-row refit. It does not constitute a complete factorial experiment: there is no screening-only arm, and the number of candidate refinements differs. The native-order arm retains native order selection but uses the common refinement; it is not the original 24-row native baseline.

Table A3. Matched supplementary comparison. E_U is clean external RMSE in noise units; UM is unmatched estimated mass as a percentage of all estimated mass. Recovery requires both order and all rates within the stated log-rate tolerance.

Path	Arm	Leakage (%)	UM (%)	E_U	$W_{1,h}$	Recovery
RAI	Native order + common refit	6.922	9.651	0.31259	0.18531	9/24
RAI	Prediction only	4.373	8.735	0.25994	0.15224	12/24
RAI	Screen + prediction	1.744	7.594	0.23236	0.11246	14/24
DOME	Native order + common refit	4.373	8.735	0.25994	0.15224	12/24
DOME	Prediction only	4.373	8.735	0.25994	0.15224	12/24
DOME	Screen + prediction	1.744	7.594	0.23236	0.11246	14/24

The incremental screened-candidate comparison reduces leakage and prediction error, while the proposal paths again produce equivalent full-method results to displayed precision. This supports a bounded contribution of the shared selection procedure, not an independent superiority claim for either path.

Appendix F.2. Supplementary Distortion Sensitivity

Twelve frozen underlying truths were subjected to four prespecified stresses: spectral AR(1) noise with correlation 0.6 and unchanged marginal variance; an unattenuated offset of one background-noise standard deviation; independent signed 8σ impulses in 0.5% of cells; and physical diffusion weighting equal to 1.05 times the supplied b values. Masks, partitions and solver settings were held fixed. These perturbations were not calibrated to an instrument and do not replace experimental mixture validation.

The offset substantially degrades recovery, despite all fits returning numerical success. For miscalibrated b, a compensating rate scale can retain accurate prediction while biasing physical diffusion; the 0.25 log-rate matching tolerance does not exclude a 5% bias. These results do not support unqualified robustness or chemical-identification claims.

Table A4. Supplementary sensitivity of RAI-S and DOME-S, whose results agree to displayed precision. Each row contains the same 12 underlying chemical truths; the correlated-noise row uses newly drawn errors. UM denotes unmatched estimated mass as a percentage of all estimated mass.

Condition	Leakage (%)	UM (%)	E_U	$W_{1,h}$	Recovery
Original iid	1.544	8.425	0.22643	0.11689	7/12
Spectral AR(1)	1.731	13.034	0.23729	0.13763	7/12
Constant offset	7.212	17.018	1.01092	0.26446	3/12
Impulsive contamination	1.872	8.547	0.26445	0.12324	7/12
b calibration +5%	1.351	7.144	0.22596	0.14808	8/12

Appendix F.3. Signal-Mask Design

The null, ablation and distortion panels reported no numerical failures. The separate discovery branch uses the first six low-b fitting acquisitions, excludes internal validation acquisitions, thresholds their mean at $4\sigma/\sqrt{6}$ and dilates the resulting mask by two spectral channels. The signal-present matched comparison and distortion panel retain masks estimated from all outer training rows, including the internal validation rows. Convergence does not imply a correct component order, spectral support or diffusion scale.

Appendix G. Extended Validation and Industrial Resin Data

These studies distinguish the independent inputs from repeated analyses of the same input. Synthetic diffusion rates and spectral amplitudes were reserved for evaluation. The controlled TRAIn-MF study used the generating factor count to isolate numerical optimization; held-out acquisitions were excluded from fitting and selection. The experimental studies report prediction separately from any independently supported chemical or diffusion assignments.

Appendix G.1. Controlled TRAIn-MF Convergence and Runtime Curvature Checks

We tested the regularized TRAIn-MF objective in Equation (29), checking the conditional curvature criterion during optimization. Nine synthetic matrices cross one, two and three finite-width shared profiles with SNR 50, 100 and 200. Each contains 32 acquisitions and 24 signal-bearing spectral channels with correlated amplitudes and overlapping peaks. The fitted factor count is fixed to the generating count to isolate optimization; this experiment does not validate automatic order selection. All 256 diffusion nodes span $[0.1, 15] \times 10^{-9}\,\mathrm{m^2\,s^{-1}}$. The log-profile width is 0.14, with generating centers $(0.7, 2.3, 7.1) \times 10^{-9}\,\mathrm{m^2\,s^{-1}}$ used in order as the factor count increases. The acquisition design is $b_i = 2.2 \times 10^9 (i/31)^{1.7}$ for $i = 0, \ldots, 31$; every fourth row is reserved for external

prediction, leaving 24 fitting rows. Independent Gaussian noise has $\sigma = 1/\mathrm{SNR}$ after normalizing the maximum unattenuated signal to one. The observations, two initialization seeds, penalties and tolerances are held fixed across alternating warm-start caps of 1, 4, 8 and 16. Every arm has the same subsequent 250-iteration SQP cap, with $\lambda_S = \lambda_A = 10^{-8}$, scaled KKT tolerance 10^{-6}, objective tolerance 10^{-9}, and inner-solver tolerance 10^{-10} with a 2000-iteration cap. An early warm stop is permitted only by the objective-and-KKT rule. A start already meeting the KKT criterion may omit SQP. Numerical failures remain in the reported endpoints.

For the interior second-difference operator with $\lambda_S > 0$, we checked $Kv \neq 0$ and full row rank of A before each profile update and at each reported SQP iterate, using a relative tolerance of 10^{-10}. If this criterion was numerically unresolved before a profile update, the subproblem used an interior-point constrained least-squares solver with the same objective and constraints. Feasibility, descent and KKT checks still applied. Support for the condition at finitely many iterates gives neither a uniform coercivity bound nor a convergence certificate. The diagnostic was tested with independent and duplicated amplitude rows, a constant kernel and zero amplitudes. In a rank-one control with eight warm sweeps, adding the diagnostic left the predicted observations unchanged.

Table A5. Controlled warm-start comparison. Warm-only stationarity counts both starts; final stationarity concerns the selected fit. CPU times are medians per two-start fit, including instrumentation. The maximum final KKT and median clean external prediction error include unconverged outputs.

Warm Cap	Pre-SQP Pass	Final Pass	CPU (s)	Max KKT	RMSE/σ
1	0/18	3/9	152.0	4.67×10^{-5}	0.2740
4	0/18	3/9	174.0	5.15×10^{-5}	0.2742
8	0/18	4/9	214.1	4.23×10^{-5}	0.2737
16	1/18	4/9	274.1	6.16×10^{-5}	0.2747

There is one realization per rank/SNR combination. This design examines optimization within a fixed budget; it does not prove convergence of the factors or identify physical constituents. Increasing the warm-start budget alone is therefore insufficient evidence that the algorithm has converged.

Only 1/72 starts meet the KKT tolerance before SQP. After refinement, the selected fits pass in 3/9, 3/9, 4/9, 4/9 cases for warm caps 1, 4, 8 and 16; 48/72 starts reach the SQP iteration cap. SQP runs in 71/72 starts, with the remaining start already satisfying the original KKT rule. On the nine paired inputs, predictions from the warm-eight and warm-sixteen fits differ by a median 0.0018 and a maximum 0.0104 noise units on external rows. Their largest absolute relative objective difference is 0.0295%. These small differences show predictive stability on this panel, but do not override the failed stationarity checks. Neither eight nor sixteen warm sweeps guarantees a stationary final fit within the SQP budget.

Of 14,292 curvature evaluations, four were numerically unresolved and triggered the interior-point fallback. All four occurred at the second initialization of the rank-three, SNR-200 input, once for each warm-start cap. They are repeated checks on one matrix, not four independent observations. The final curvature condition held numerically for all 72 starts, although many selected fits did not meet the stationarity criterion. Conditional curvature and stationarity of the full estimate thus describe different properties.

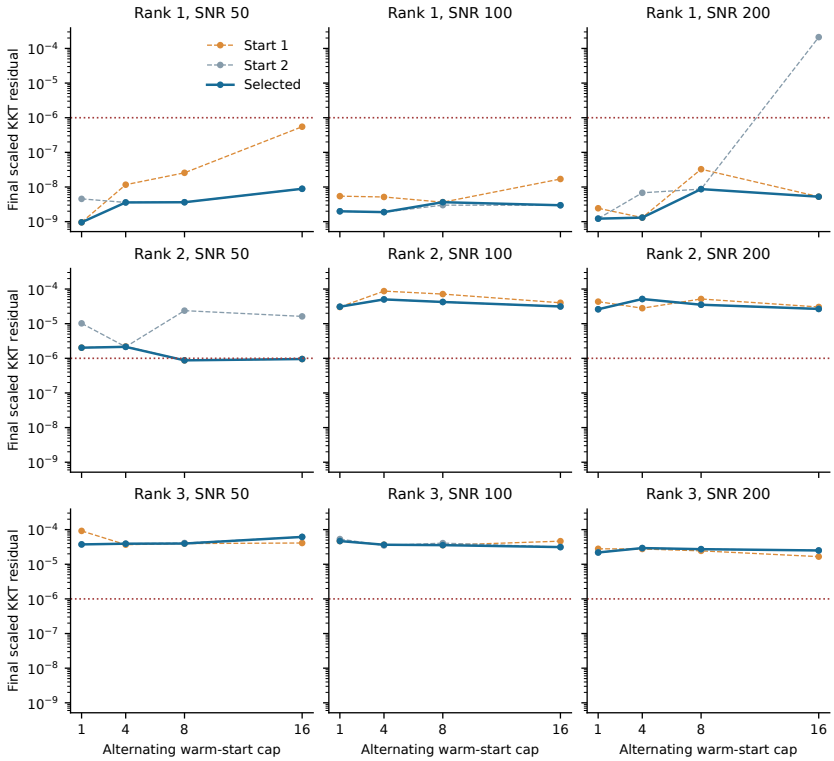


Figure A1. Controlled TRAIn-MF warm-start sensitivity with a common subsequent SQP budget. Numerical stationarity is assessed by the declared independent KKT criterion; all unresolved outcomes are retained. Repeated budgets on one rank/SNR input are paired conditions rather than new independent samples. Orange dashed and blue-gray dashed curves denote the first and second initializations, respectively; the solid blue curve denotes the selected fit. The red dotted line marks the KKT tolerance of 10^{-6}.

Appendix G.2. Factorial Separation of the Shared Selection Stages

We added a paired 2×2 intervention on the 36 signal-present atomic problems, with a common proposal bank per method constructed from 18 fitting rows, six reserved rows and the same final 24-row refit. $S = 1$ forces the conditional spectral mask and $S = 0$ retains unrestricted amplitudes. $P = 0$ retains the native-selected nominal proposal order, whereas $P = 1$ applies predictive order selection. Unsupported rates may disappear during refinement, so the nominal proposal order is not necessarily the final active order. The separate production reference permits both screened and unrestricted candidates and reproduces the published selector; it is not a factorial cell.

All 360 final outputs, including the separate production references, passed their numerical checks; no selected candidate failed its numerical criterion. The leakage interaction $(S1P1 - S0P1) - (S1P0 - S0P0)$ is -1.511 percentage points for RAI (paired problem-bootstrap 95% interval $[-2.603, -0.598]$) and -0.763 for DOME ($[-1.670, -0.186]$). These descriptive intervals do not establish a universal interaction or provide multiplicity-adjusted inference. The two final proposal paths remain effectively equivalent. DOME's unmatched mass increases slightly from $S1P0$ to $S1P1$ despite lower average leakage and prediction error. Original signal masks are retained, including their original internal validation reuse; this is a matched mechanism comparison, not a new calibration experiment. Per-arm logical cost accounts for shared candidate fits and does not represent separately measured end-to-end timing. In the weak-component problem `test_011_weak`, forced screening and prediction return three rates but match only two true rates; the minority rate remains unrecovered. Its lower E_U and $W_{1,h}$ therefore do not establish recovery of the

additional species. Correct order rises from 25 to 26 problems relative to production choice, while complete recovery remains 23/36 for both.

Table A6. Matched factorial experiment. Leakage and unmatched mass (UM) are percentages. Recovery requires both order and log-rate matching.

Path	Cell	Leakage	UM	E_U	$W_{1,h}$	Recovery
RAI	$S0P0$	6.092	8.627	0.30677	0.16805	15/36
RAI	$S1P0$	5.301	8.627	0.29087	0.15092	15/36
RAI	$S0P1$	3.952	8.016	0.26189	0.14512	20/36
RAI	$S1P1$	1.650	7.265	0.23145	0.10767	23/36
DOME	$S0P0$	3.952	8.016	0.26189	0.14512	20/36
DOME	$S1P0$	2.413	7.257	0.23856	0.11588	21/36
DOME	$S0P1$	3.952	8.016	0.26189	0.14512	20/36
DOME	$S1P1$	1.650	7.265	0.23145	0.10767	23/36
Both	Production	1.728	7.257	0.23299	0.10944	23/36

Appendix G.3. Adaptive Weak-Component Recovery

Ninety-six new problems span four generating minority-amplitude factors: 0.02, 0.04, 0.08 and 0.16. Three SNRs (50, 100 and 200), with eight independent truth/noise realizations per cell, complete the design. Each factor multiplies the third spectral template before the total unattenuated signal is normalized; it is not a molar fraction. The mask is formed from fitting acquisitions only, excluding both internal validation and external test rows. The frozen adaptive selector and rate-matching tolerance remain unchanged.

All 192 method fits pass their numerical criteria. RAI-S matches the minority rate in 45 of 96 problems and DOME-S in 44 of 96; these are paired method results, not 192 independent chemical mixtures. At factor 0.02 the counts across increasing SNR are $(0, 0, 1)/8$ for both methods. At factor 0.08 they are $(1, 8, 8)/8$. The intermediate factor 0.04 gives $(0, 1, 4)/8$ for RAI-S and $(0, 0, 4)/8$ for DOME-S. These visible failures prevent a general claim of reliable weak-component discovery. Masking also contributes: mean retained minority spectral mass varies from about 0.59 at the weakest low-SNR cell to 0.99 at the strongest high-SNR cell. A channel outside the mask is unestimated.

Exact two-sided 95% Clopper–Pearson intervals are retained per cell in the machine-readable supplement; for example, zero of eight gives an upper endpoint 0.369 and eight of eight a lower endpoint 0.631. They are descriptive cell-wise intervals without multiplicity adjustment. Aggregate counts across the prespecified SNR/factor mixture are not a universal power guarantee.

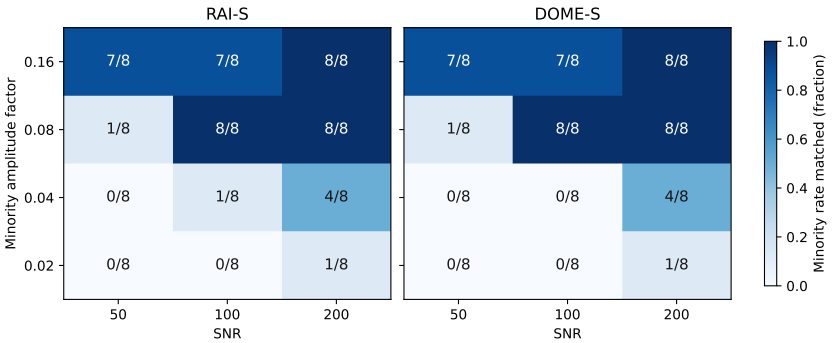


Figure A2. New weak-component panel. Each cell counts matched minority rates in eight independent problems, using the original log-rate tolerance 0.25. The methods use paired inputs. These counts estimate recovery under the declared generator, not a calibrated probability of chemical presence.

Appendix G.4. Expanded Neural Policy Evaluation

We trained five CPU networks with different seeds using the same 80-problem trace bank. Each run separates 64 training from 16 validation problems and uses training records alone to normalize features. The new test panel contains 120 problems: 72 in the original physical domain, 24 at unseen SNRs (25, 35 and 500), and 24 with unseen acquisition exponents and diffusion weightings. The five policies and the deterministic proposal-order control share the six-round cap, a shortlist of one with fallback, and the same numerical verifier. The control is the classical shortlist variant; it differs from both the exhaustive controller and the atomic RAI estimator.

Table A7. Independent neural evaluation. $W_{1,h}$ averages 105 signal-present problems; E_U includes all 120 problems, including 15 null controls. Numerical success is a solver criterion. Times are mean solver seconds and exclude checkpoint loading and serialization.

Method	$W_{1,h}$	E_U	Numerical Success	Seconds	Tree Fits
Classical shortlist	0.33383	0.29961	120/120	22.03	20.76
RAI-Net 172901	0.33341	0.29956	120/120	17.44	19.24
RAI-Net 172902	0.33319	0.29941	120/120	18.91	19.66
RAI-Net 172903	0.33323	0.29952	120/120	18.13	19.49
RAI-Net 172904	0.33390	0.29951	120/120	18.87	18.94
RAI-Net 172905	0.33271	0.29944	120/120	19.15	19.55

The seed-average neural-minus-classical $W_{1,h}$ difference is -0.00054 (paired problem-bootstrap 95% interval $[-0.00174, 0.00056]$); the corresponding E_U difference is -0.00012 ($[-0.00044, 0.00010]$). The pooled bootstrap resamples whole problems within each domain, preserving domain counts; the five seed outputs on a problem are paired, not independent tests. Individual seed mean $W_{1,h}$ differences range from -0.00111 to 0.00007. ID: $\Delta W_1 = -0.00136$, interval $[-0.00294, -0.00003]$ ($n = 63$); individual seed differences range from -0.00165 to -0.00076. OOD_noise: $\Delta W_1 = -0.00059$, interval $[-0.00230, 0.00070]$ ($n = 21$); individual seed differences range from -0.00142 to 0.00013. OOD_acquisition: $\Delta W_1 = 0.00197$, interval $[-0.00135, 0.00559]$ ($n = 21$); individual seed differences range from -0.00106 to 0.00430. All 720 final method outputs pass their numerical checks, with 0 execution errors and 0 failed final checks. The analysis retains failed numerical outputs whenever they occur, rather than removing them from the averages. Intermediate failures still occur: 98/2371 control subproblems and 476/11026 neural subproblems fail the search-stage checks, with 91–99 failures per policy. Each method encounters at least one failed search fit in 48/120 problems. The selected training paths and all final refits nevertheless pass their checks. These counts describe solver events and should not be treated as independent statistical trials. Mean differences from the classical control are -3.524 s per problem for solver time and -1.382 for the number of fixed-tree fits (neural minus classical). Equal search caps therefore do not imply equal realized effort. Training the five policies took 3.455 s after data loading. Generating the common training traces took 1690.7 s of summed computation; that sum cannot be compared directly with wall-clock time. These measurements do not establish an end-to-end speedup. Four selected checkpoints reach the 400-epoch budget, so learning convergence is also unresolved. The study supports neither a universal accuracy advantage nor experimental chemical identification.

An exploratory breakdown by problem family finds lower mean shape error for all five policies on the 15 broad-profile problems, but higher error for four of the five policies on the 90 atomic problems. We did not use these patterns to select a seed or tune the models. They are post hoc observations, not independent confirmatory tests.

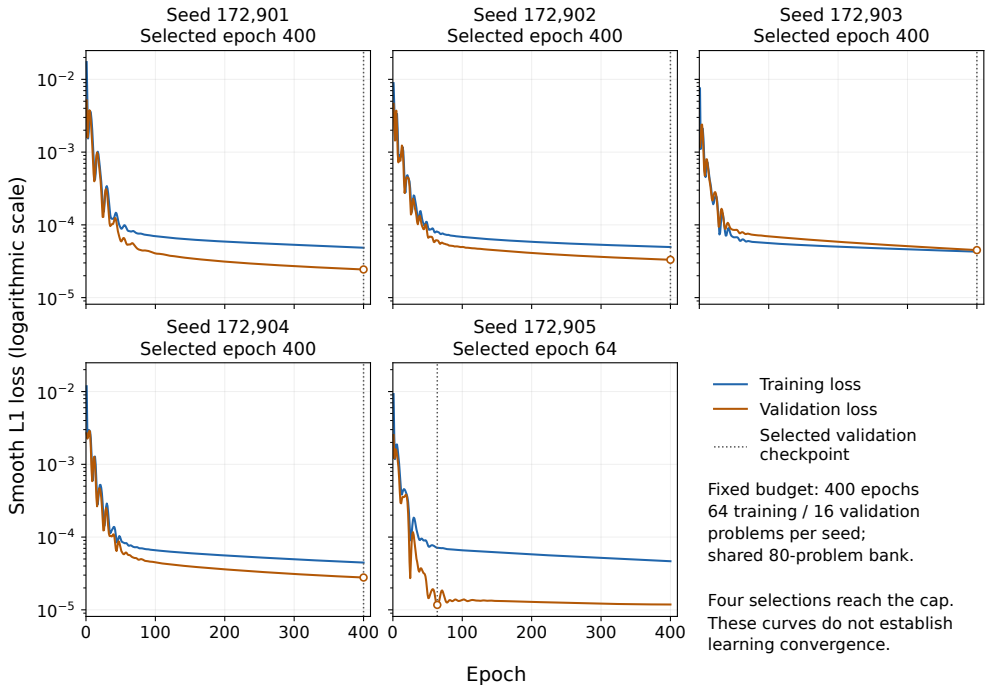


Figure A3. Training and whole-problem validation Smooth L1 loss for five separately seeded RAI-Net rankers. Each run uses 64 training and 16 validation problems from the same historical 80-problem bank. Dotted lines and open markers identify the lowest observed validation loss checkpoint. Four selections reach the declared 400-epoch budget, so these histories do not establish learning convergence. The historical trainer records training loss before the Adam update and validation loss after it. Test observations are absent from normalization, training and checkpoint selection.

Appendix G.5. Shared External-Reference Comparison

The additional comparison uses the same frozen observations, masks and outer test acquisitions. The physical interval is common to grid methods and positive measure eligibility. Native continuous fits are retained with their original rate constraints rather than silently projected into that interval. Matrix zero served as an adapter pilot and is retained in the raw outputs but excluded symmetrically from the iid summary, leaving 47 matrices (41 signal-present and six null). Twelve underlying signal-present problems also receive each of the baseline and correlated-noise stresses. These are retrospective comparisons, not a new untouched synthetic population. Outer test rows are not used in the internal parameter or order decisions.

SCORE uses the attributed GNAT translation, signed conditional least squares and its original initialization, with a declared reserved-acquisition wrapper selecting orders zero through four. Candidate selection enforces interval eligibility, but the final continuous refit is unconstrained in log rate; out-of-interval results are retained and flagged. The positive-amplitude sensitivity uses the same wrapper and proposal budget. LOCODOSY retains its native local SVD order and reduction rules on the common contiguous signal windows; its sum of local components is not a global molecular count. Its native rate domain has zero as the lower limit and the common upper limit. LOCODOSY uses the training observations for its order rule; only the validation-selected adapters use the internal split. Five of the 47 iid LOCODOSY outputs contain rates outside the declared positive measure interval; none is projected or discarded. No outer test row participates in these decisions. Positive Tikhonov selects its curvature weight from $\{0.01, 1, 100, 10^4\}$ on internal validation. The PALMA constrained objective uses a training-only positive low-b scale, entropy weights $\{0.01, 0.1, 0.5\}$ selected on internal validation, a declared 95% Gaussian noise-ball radius and 5000 final PPXA+ iterations. A batched Woodbury graph projection was checked

against the reviewed scalar QR engine; this is an implementation of the stated objective, not a claim of identical performance to all upstream application presets.

Table A8. Common external-reference comparison. E_U is clean external error in noise units over signal-present cases. Numerical success uses each method's own criterion; all returned outputs enter E_U, including nonconverged ones. The column *n* counts only outputs admitting a nonnegative measure within the declared interval and is therefore a different population for signed methods.

Method	Numerical Success	E_U	$W_{1,h}$	n
RAI-S	47/47	0.23477	0.1201	41
DOME-S	47/47	0.23477	0.1201	41
SCORE-CV (signed)	47/47	0.30329	0.0000	3
SCORE-NNLS-CV	47/47	0.26681	0.1563	41
LOCODOSY (signed)	47/47	0.26114	0.0013	4
Positive Tikhonov	47/47	0.33523	0.3057	41
PALMA objective + CV	0/47	0.73627	0.4225	41

The table compares prediction under the stated fitting conditions, rather than chemical identification. Signed cancellation can reduce error without yielding a positive component representation. Similarly, PPXA+ has not reached the required optimality criterion for the PALMA objective, so its finite-budget result cannot rank the fully optimized method. The supplement retains all convergence flags and curve-level constraint diagnostics. CONTIN, UPEN, DRECT and DREAM were not run under this common protocol; Table 1 states what those comparisons would require. Positive Tikhonov is a separate control and is not presented as the full CONTIN procedure. The executed comparisons cover the closest joint-component references and two positive distributional controls.

On the twelve baseline-stress cases, positive SCORE has a slightly smaller mean clean error than RAI-S (1.00044 versus 1.01092 noise units); the paired descriptive interval for the difference is $[-0.02583, 0.00196]$ and includes zero. There is no demonstrated robust advantage on this offset panel. Correlated-noise results and all paired contrasts are included in the machine-readable supplement; they are not independent replications of the underlying twelve signal configurations.

Appendix G.6. Experimental Prediction on Industrial Resin Spectra

We analysed spectra from four industrial resin specimens identified as TP194-2623086, TP194-2623088, TP194-2623084 and TP194-2619082. These are specimen identifiers; their digits are retained without thousands separators. Each series contains three blocks at diffusion times 75, 100 and 125 ms, with 23 acquisitions and 65,536 spectral channels per block. The twelve blocks belong to four specimens and are not independent mixture preparations. We fit all 17,386–21,314 channels per block within the detected positive absorption support, preserving their measured signs. Five acquisitions are withheld for prediction, and support detection also excludes the internal validation rows. A single scalar normalizes each matrix; channels are neither rescaled individually nor replaced by integrated bands or a thinned fit grid. Outside the support, signal remains unestimated. The full-spectrum check detects 75–219 strongly negative points per block, including 53–187 outside the positive support. We apply no automatic phase, baseline, absolute-value or clipping correction.

RAI-S and DOME-S use their frozen support-aware estimators with a four-atom search cap. SCORE-CV uses the attributed SCORE implementation with signed amplitudes and native initialization, augmented by internal acquisition-based order selection over one to four atoms [35]. SCORE-NNLS-CV changes the conditional amplitude problem to NNLS while retaining the same initialization, optimizer budget and selection wrapper; it is explicitly a constrained variant, not the original signed SCORE method. LOCODOSY-SCORE retains the native local singular-value order heuristic and reduction rules [37], using the same detected windows. Its output rows are local contributions, not a global species count. Small-rank batched NNLS is checked against the native per-channel solver; the RAI objective, gradient and predictions are also checked against the original implementation. All selected channels contribute to every reported error.

Table A9. Experimental acquisition holdout prediction on twelve blocks, grouped into four industrial resin specimens. NRMSE divides held-out RMSE by the training RMS on the same support. NC counts final global optimizer nonconvergence; unstable flags denote kernel condition above 10^8 or amplitude expansion above 10^4; cap counts selection of the maximum four global atoms. These flags are not exclusions. Dashes denote quantities that are not global estimands for LOCODOSY.

Method	Mean NRMSE (%)	Median (%)	NC	Unstable	Cap
RAI-S	1.1226	1.1119	0	0	12
DOME-S	1.1226	1.1119	0	0	12
SCORE-CV	0.7740	0.7884	6	8	8
SCORE-NNLS-CV	1.1230	1.1119	0	0	12
LOCODOSY-SCORE	0.8914	0.8620	–	–	–

All 60 of 60 planned predictions were completed. RAI-S and DOME-S reach the cap and trigger the atomic model mismatch warning in all twelve blocks. Their largest relative prediction difference is 1.41×10^{-12}, providing no independent predictive advantage for either proposal mechanism. Reaching the cap also does not establish four chemical species. Positive SCORE serves as a constraint control. Signed SCORE predicts better in several blocks, but 6 of its twelve final fits fail to converge within the budget and 8 have unstable global amplitudes. We retain these predictions in the table. LOCODOSY covers 174 windows, with 0 nonconverged local fits and 0 failed windows. Low prediction error from canceling signed amplitudes alone cannot establish reliable identification.

Background neighboring-channel correlations range from 0.999198 to 0.999757. Consequently, the background MAD scale is descriptive, and these processed data do not validate an independent-Gaussian interpretation of support scores or channel-wise confidence intervals. Acquisition-order effects and residual phase/baseline distortion are not separately identified. Spectral-factor matching across the three diffusion times is reported as a descriptive consistency check, not as chemically verified correspondence.

The same HDO-based calibration was used at all three diffusion times. If $D = 10^{-9} c\kappa$, where c is the calibration coefficient, first-order uncertainty propagation would require

$$\mathrm{Var}(D) \simeq 10^{-18} \left[\kappa^2 \,\mathrm{Var}(c) + c^2 \,\mathrm{Var}(\kappa) + 2c\kappa \,\mathrm{Cov}(c, \kappa) \right]. \tag{A20}$$

The reference and thermal-model contributions to $\mathrm{Var}(c)$ are not quantified in the source calibration; they are not set to zero. The maps are conditional on that calibration, and no absolute diffusion-accuracy interval or molecular-weight estimate is asserted.

The available records do not link these four identifiers to a verified composition. We therefore assign no compositions or diffusion values from other batches to these specimens.

The measurements provide evidence about prediction on industrial resins, but not a test against known component counts or independently measured component diffusion coefficients. This is especially relevant when finite atomic models are fitted to potentially polydisperse materials.

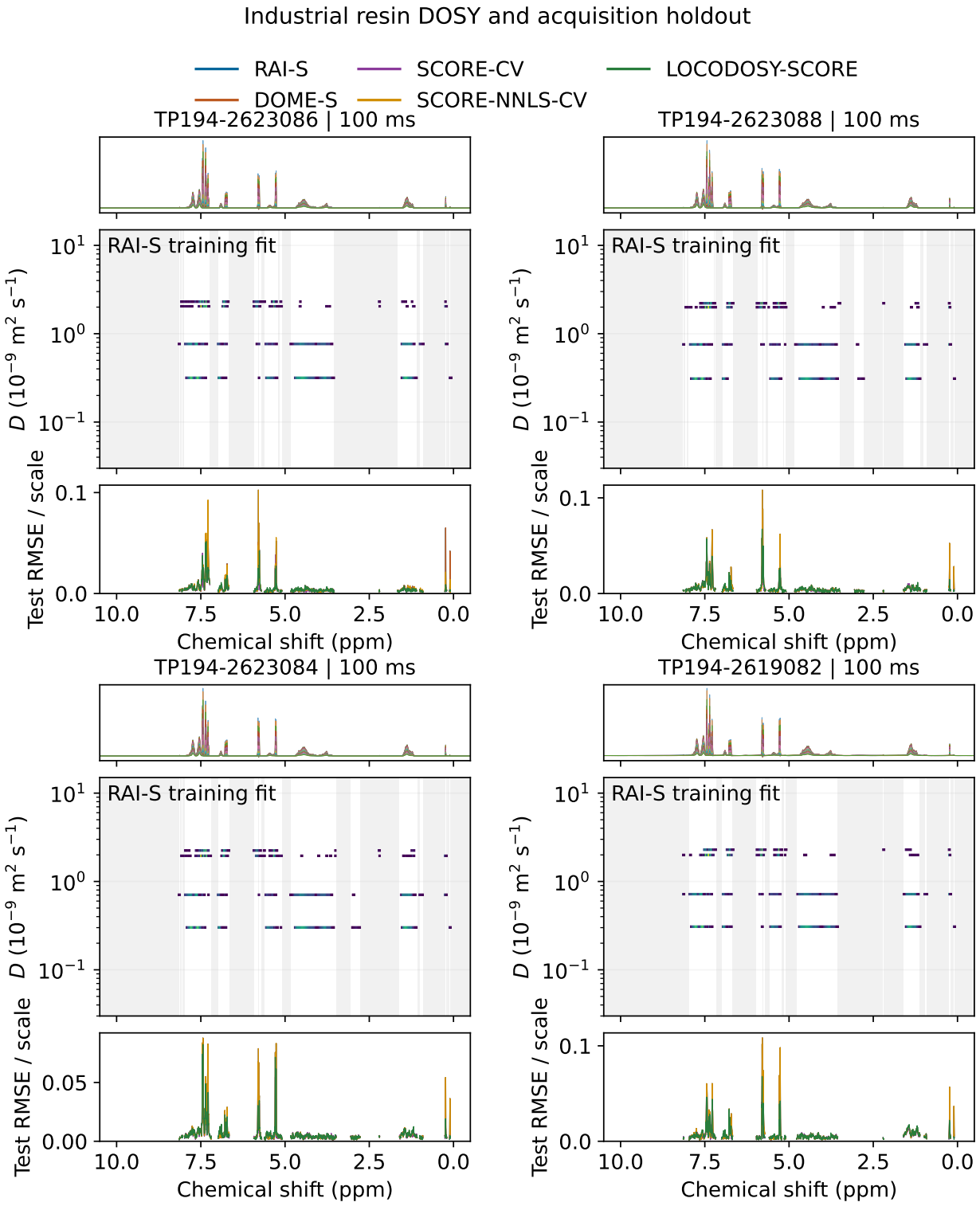


Figure A4. Industrial resin spectra at 100 ms: measured spectral families (top), RAI-S training-fit atomic DOSY maps (middle) and held-out error profiles for the five estimators (bottom). All detected support channels enter the fits; gray map regions are unestimated. Display thinning is confined to traces and error curves. Atoms are shown without fabricated distributional widths, and no chemical ground-truth map is available. The TP194 labels are specimen identifiers, not measured quantities.

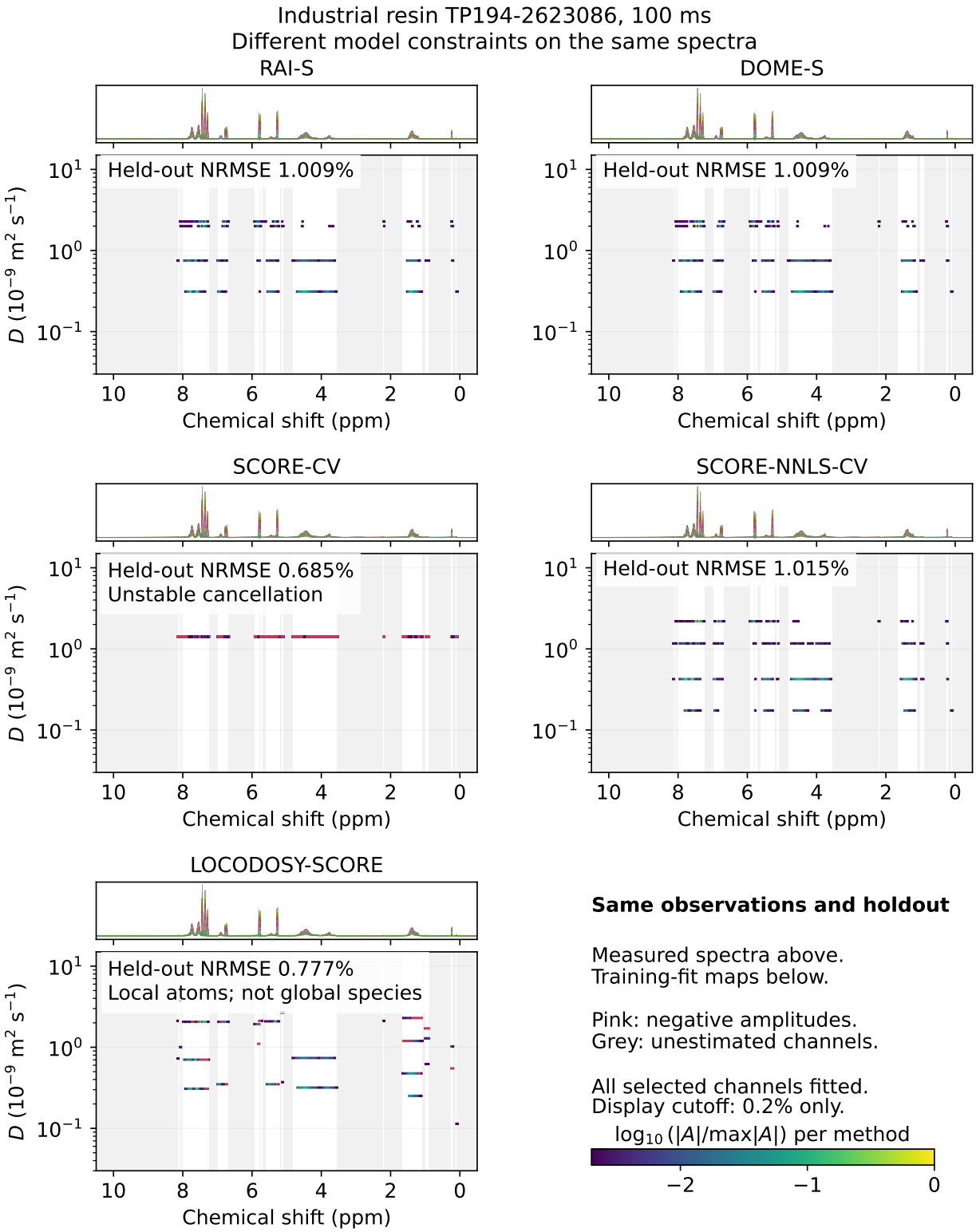


Figure A5. The five estimators on industrial resin TP194-2623086 at 100 ms, using identical observations and acquisition holdout. Colors show relative absolute amplitude within each method; negative amplitudes are pink and unestimated regions are gray. The 0.2% display cutoff does not affect fitting or scoring. The nearly coincident signed SCORE atoms exhibit large cancellation despite a lower held-out prediction error. LOCODOSY lines are local contributions. TP194-2623086 is a specimen identifier and is therefore written without thousands separators.

Appendix G.7. A Public Multicomponent DOSY Example

We additionally processed the public `PropPentSuc_DOSY.fid` example distributed by the official DOSY Toolbox downloads page [51]. Its internal sample text documents propanol and sucrose in D_2O; its directory name is consistent with the 1-propanol/1-pentanol/sucrose example illustrated in the GNAT manual [52]. The latter does not explicitly identify this archived file, so pentanol is not treated as a verified additional label. Neither a certified complete composition nor independently measured diffusion coefficients accompany the example used here. It therefore supplies a public multicomponent experimental check, not known-order or absolute-D ground truth.

The archive contains 30 acquired gradient levels. Acquisitions 4, 8, 12, 16, 20, 24 and 28 were reserved for external prediction; five of the remaining acquisitions formed the internal validation split. A single common linear ACME phase [53] was estimated from the other 18 discovery acquisitions after Fourier transformation and then frozen for all methods and acquisitions. The archived phase alone had left visible dispersion in discovery-only quality control. We applied no baseline subtraction, absolute-value rectification or spectral alignment. All 4074 original spectral channels within the discovery-detected positive absorption support were retained from the 32,768-point spectrum. Channels outside that support remain unestimated. The common order cap was four.

Following the GNAT Varian import convention, the nominal encoding was

$$b_i = \gamma^2 T_{\mathrm{enc}} \left(g_{i,\mathrm{DAC}} c_{\mathrm{DAC}}\, 10^{-2} \right)^2, \tag{A21}$$

where T_{enc} is the archived `dosytimecubed` parameter and c_{DAC} is the archived conversion to $\mathrm{G\,cm^{-1}}$. The same homogeneous gradient kernel was used for every method. No independent gradient calibration could be matched to this acquisition, so no further gradient correction was applied. Nonuniform gradients and other unmodeled acquisition effects therefore remain possible explanations for residual curvature; the nominal D scale is not an independently calibrated accuracy claim.

Table A10. One public experimental dataset, shared support and acquisition holdout. Numerical convergence and chemical identification are distinct.

Method	Test NRMSE (%)	Atoms	Converged
RAI-S	0.582	4	yes
DOME-S	0.582	4	yes
SCORE-CV	0.232	4	yes
SCORE-NNLS-CV	0.582	4	yes
LOCODOSY-SCORE	0.270	31 *	local

* Sum of local contributions, not a global molecular count. All local solver calls converged.

RAI-S and DOME-S returned effectively identical predictions (relative difference 7.6×10^{-11}), both reaching the cap and activating the atomic model mismatch diagnostic. The positive-amplitude SCORE variant agreed closely with them. Signed SCORE had the lowest prediction error; its negative-amplitude fraction was 9.54% of total absolute amplitude. Unlike the unstable signed fits in some resin blocks, this fit converged and was not flagged by the declared kernel conditioning or amplitude expansion checks. Thus, its predictive advantage cannot be dismissed as numerical failure, although a signed decomposition does not establish nonnegative constituent spectra. LOCODOSY also predicted better than the positive global fits. These results do not demonstrate superiority of RAI-S or DOME-S over the references, or an independent benefit of their proposal mechanisms. The single archived experiment, correlated processed noise (adjacent background correlation 0.983), unknown full composition and nominal gradient calibration preclude a population-

level detection claim or certified chemical assignments. Solvent/reference resonances were not silently removed, and an atomic row is not equated with a molecule.

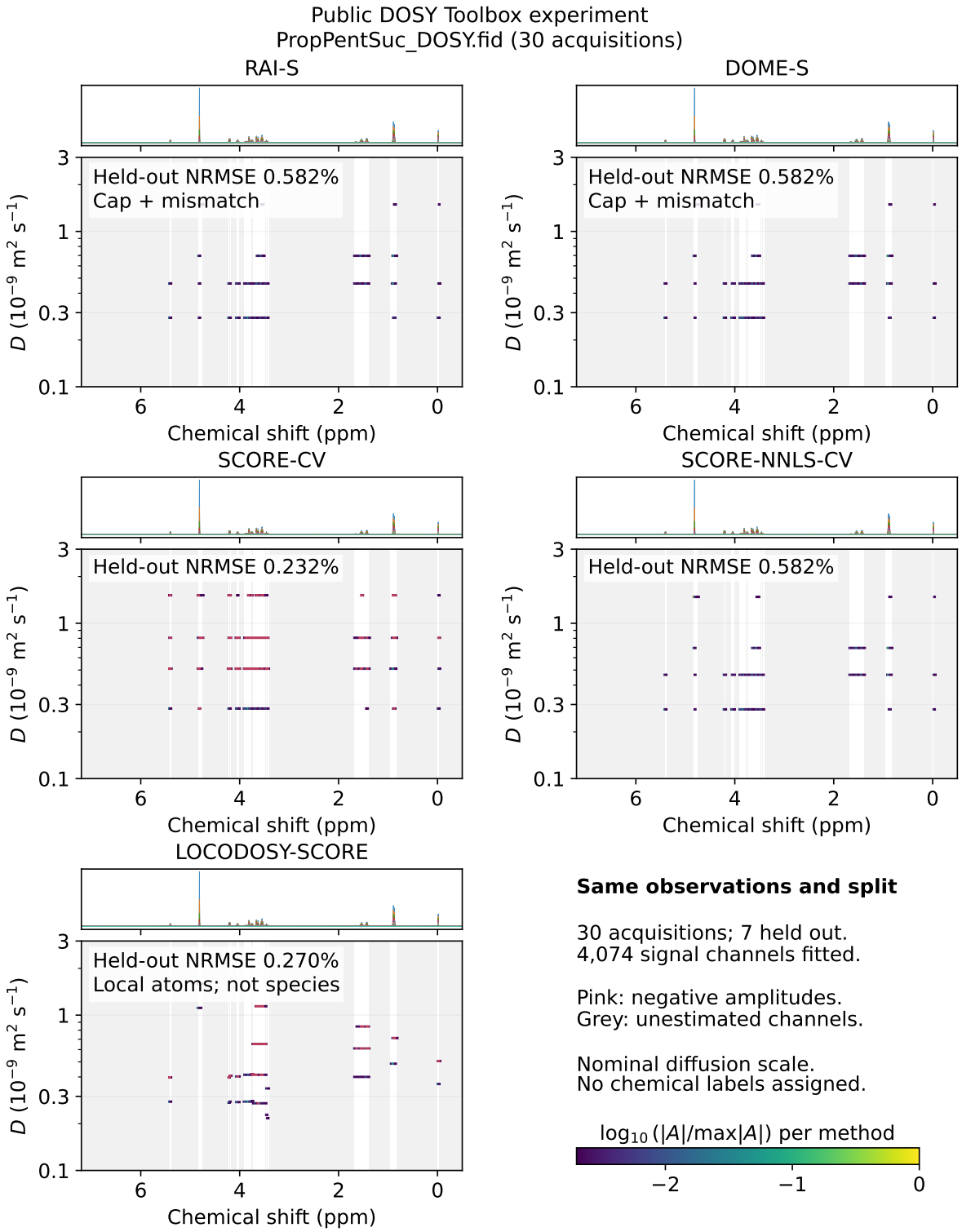


Figure A6. Public DOSY Toolbox example. Measured spectral families appear above the training-only fitted atomic maps. Every selected spectral channel is fitted; only the display of the upper traces is decimated. Pink denotes signed negative amplitudes, gray denotes unestimated channels, and the 0.2% relative display cutoff does not prune the fit. Atomic rows have no inferred diffusion width or assigned chemical identity.

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